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# Academic Explanation of the Poisson + CBF + Contingency System

From perception and occupancy maps to safety filters and multi-zone landing

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A detailed study document based on the supplied implementations and experimental outputs

## Two case studies explained independently

1. Static image  $\rightarrow$  2-D occupancy  $\rightarrow$  Poisson field  $\rightarrow$  velocity-level CBF.
2. 3-D world  $\rightarrow$  Poisson field  $\rightarrow$  acceleration-level HOCBF  $\rightarrow$   $r$ -out-of- $p$  contingency and safe landing-zone switching.

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# Abstract

This document develops, from first principles and with explicit mathematical detail, a robotics safety architecture that combines three complementary ideas. First, complex discrete geometry is converted into a smooth safety function by solving Poisson’s equation. Second, Control Barrier Functions (CBFs) and High-Order Control Barrier Functions (HOCBFs) convert that function into affine constraints on velocity or acceleration. Third, a combinatorial contingency filter allows the vehicle to pursue an active landing zone while preserving the estimated viability of at least  $r$  out of  $p$  candidate zones.

The presentation separates mathematical theory, numerical implementation, and empirical evidence. It states which assumptions support continuous-time invariance, which operations are discrete approximations, how the Poisson linear system is verified, and which claims cannot yet be transferred to a real vehicle. Two reproducible examples are analyzed independently using their generated figures and configuration data: a 2-D overhead-image experiment and a 3-D contingency-aware landing prototype with target switching.

## Scope of the claim

The second example uses smooth certificates  $\rho_i = c_i - W_i$  inspired by reachability, but it does **not** use Hamilton–Jacobi reach–avoid value functions. Preserving  $\rho_i \geq 0$  certifies the set defined by these numerical proxies within the reduced-order model. By itself, it is not a full landing guarantee for PX4, a complete rotorcraft model, real sensors, or real actuators.

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# Notation Guide

|                      |                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------|
| $O$                  | Boolean occupancy matrix. In the code, <code>True</code> means occupied.                              |
| $\Omega$             | Free domain on which Poisson’s equation is solved.                                                    |
| $\partial\Omega$     | Dirichlet frontier: free cells adjacent to occupancy and, when configured, the outer domain boundary. |
| $f_P$                | Poisson forcing function; it shapes the interior geometry of the field.                               |
| $h$                  | Poisson safety function.                                                                              |
| $\nabla h$           | Safety gradient, used directly by velocity-level CBFs.                                                |
| $\nabla^2 h$         | Safety Hessian, required by acceleration-level HOCBFs.                                                |
| $p, v, a$            | Position, velocity, and acceleration of the reduced-order model.                                      |
| $V$                  | Control Lyapunov Function for the active stabilization task.                                          |
| $W_i$                | Smooth estimated cost of reaching landing zone $i$ in the contingency prototype.                      |
| $c_i$                | Available budget associated with landing zone $i$ .                                                   |
| $\rho_i = c_i - W_i$ | Estimated viability certificate or margin for zone $i$ .                                              |
| $\tilde{\rho}$       | The $r$ -th largest certificate, used as the $r$ -out-of- $p$ pivot.                                  |
| $\omega$             | Nonnegative auxiliary variable in the combinatorial filter.                                           |

## Part I

# Mathematical and Computational Foundations

# System Overview

## 1.1 Why geometry, safety, and mission planning are separated

An autonomous system must answer at least three distinct questions:

1. **Geometry:** where are the occupied regions, and what is the boundary of free space?
2. **Dynamic safety:** which control commands keep the robot inside the safe set?
3. **Mission and contingency:** which target should be pursued, and which alternatives must remain available?

The architecture studied here does not collapse these questions into one heuristic potential. The map is first transformed into a scalar field  $h$ . That field is then converted into control constraints. Finally, a QP cost tries to preserve the nominal task command, while contingency constraints determine which landing alternatives must remain viable.

### Key idea

Poisson's equation is **not the controller**; it produces a differentiable geometric object. A CBF or HOCBF **does not choose the mission**; it filters control commands to protect a set. The contingency planner **does not replace environmental safety**; it determines which alternative targets should be preserved.

## 1.2 End-to-end data flow

The basic computational chain is

$$\begin{aligned} \text{image or 3-D world} &\longrightarrow O \longrightarrow (\Omega, \partial\Omega) \longrightarrow h, \nabla h, \nabla^2 h, \\ &\longrightarrow \text{CBF/HOCBF-QP} \longrightarrow \text{safe command}. \end{aligned} \tag{1.1}$$

For multiple landing zones, the chain is extended as

$$(h, \nabla h, \nabla^2 h, p, v) \longrightarrow W_i, c_i, \rho_i \longrightarrow \tilde{\rho} \longrightarrow r\text{-out-of-}p \text{ filter}. \tag{1.2}$$

## 1.3 Two reduced-order models

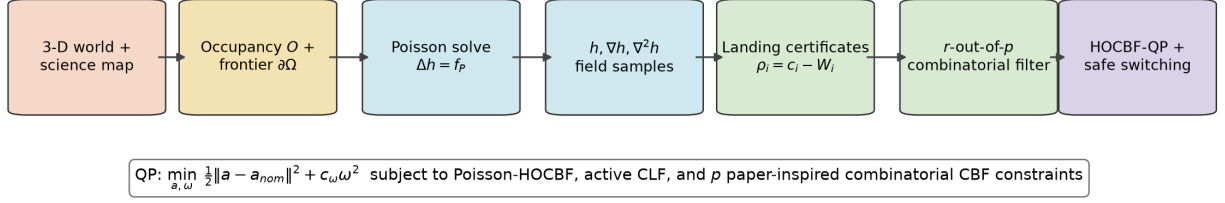
The two case studies use different dynamics.

### 1.3.1 Single integrator

For the 2-D experiment,

$$\dot{p} = u, \quad p, u \in \mathbb{R}^2. \tag{1.3}$$

Contingency-aware Poisson-CBF landing workflow (offline reduced-order prototype)



**Figure 1.1:** Complete workflow of the 3-D prototype. Each block represents a distinct data contract: geometry, PDE synthesis, field derivatives, landing certificates, combinatorial filtering, and the final QP.

The control input  $u$  is a velocity. It appears in the first derivative of  $h(p)$ , so a relative-degree-one CBF is appropriate.

### 1.3.2 Double integrator

For the 3-D experiment,

$$\dot{p} = v, \quad \dot{v} = a, \quad p, v, a \in \mathbb{R}^3. \quad (1.4)$$

The control input  $a$  is an acceleration. It does not appear in  $\dot{h}$ , but it does appear in  $\ddot{h}$ ; therefore, a relative-degree-two HOCBF is required.

# From Perception to an Occupancy Matrix

## 2.1 Planar calibration and homography

A photograph can be interpreted as a metric map only when the scene is approximately planar and the camera is overhead, or when the image is rectified through a homography. Given four ordered corners of the working region, a homography  $H \in \mathbb{R}^{3 \times 3}$  maps an original image pixel  $(u, v)$  to a rectified pixel  $(\tilde{u}, \tilde{v})$ :

$$\lambda \begin{bmatrix} \tilde{u} \\ \tilde{v} \\ 1 \end{bmatrix} = H \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \quad \lambda \neq 0. \quad (2.1)$$

A rectified grid with  $N_x \times N_y$  nodes represents the physical rectangle  $[0, W] \times [0, H_w]$ , so

$$\Delta x = \frac{W}{N_x - 1}, \quad \Delta y = \frac{H_w}{N_y - 1}. \quad (2.2)$$

The division by  $N - 1$  is essential because both endpoints of the physical domain are included in the grid.

## 2.2 Classical image segmentation

In the first example, obstacle perception is based on HSV thresholds. If  $I_{HSV}(q)$  is the color vector at pixel  $q$ , the raw mask is

$$M_{\text{raw}}(q) = \begin{cases} 1, & \ell \leq I_{HSV}(q) \leq u, \\ 0, & \text{otherwise,} \end{cases} \quad (2.3)$$

where the inequalities are componentwise. The code subsequently applies a deterministic post-processing chain:

1. Gaussian blur;
2. binary thresholding;
3. morphological opening to remove small isolated noise;
4. morphological closing to fill narrow gaps;
5. removal of small connected components;
6. optional hole filling;
7. optional manual correction.

## 2.3 Boolean convention and geometric inflation

Occupancy follows the convention

$$O_{ij} = \begin{cases} 1, & \text{occupied cell,} \\ 0, & \text{free cell.} \end{cases} \quad (2.4)$$

In Python, occupied cells are represented by `True`.

A real robot is not a point. If the effective inflation radius is

$$r_{\text{inf}} = r_{\text{robot}} + r_{\text{perception}}, \quad (2.5)$$

the configuration-space obstacle set is constructed by the Minkowski sum

$$\mathcal{O}^+ = \mathcal{O} \oplus B_{r_{\text{inf}}}. \quad (2.6)$$

On an anisotropic grid, the implementation uses

$$R_x = \left\lceil \frac{r_{\text{inf}}}{\Delta x} \right\rceil, \quad R_y = \left\lceil \frac{r_{\text{inf}}}{\Delta y} \right\rceil. \quad (2.7)$$

The binary obstacle mask is dilated by an elliptical structuring element that conservatively spans these radii.

### Technical warning

Inflation is not a cosmetic visualization step. It defines the occupied set against which the simulation is evaluated. Too little inflation can produce a trajectory that is safe for a point but unsafe for the physical body. Too much inflation can close corridors and destroy feasibility.

## 2.4 Free domain and Dirichlet frontier

The discrete free domain is

$$\Omega_h = \{q : O(q) = 0\}. \quad (2.8)$$

A free cell is marked as a boundary cell if it touches occupancy under four-neighbor connectivity in 2-D or six-neighbor connectivity in 3-D. If the external edge is treated as a wall, it is included as well:

$$\partial\Omega_h = \{q \in \Omega_h : \exists q' \in \mathcal{N}(q), O(q') = 1\} \cup \partial\mathcal{W}_h. \quad (2.9)$$

The unknown PDE values live in

$$\Omega_h^\circ = \Omega_h \setminus \partial\Omega_h. \quad (2.10)$$

# Synthesizing a Safety Function with Poisson's Equation

## 3.1 Boundary-value problem

The safety function is defined as the solution of the Dirichlet problem

$$\begin{cases} \Delta h(y) = f_P(y), & y \in \Omega, \\ h(y) = 0, & y \in \partial\Omega. \end{cases} \quad (3.1)$$

It is often convenient to write

$$-\Delta h = q, \quad q = -f_P > 0, \quad (3.2)$$

which makes the positive-definite structure of the discrete matrix explicit.

### 3.1.1 Why negative forcing produces a positive interior

If  $f_P < 0$ , then  $\Delta h < 0$ , so  $h$  is superharmonic. With  $h = 0$  on the boundary, the maximum principle implies, under the usual regularity assumptions,

$$h(y) \geq 0 \quad \text{in } \Omega. \quad (3.3)$$

If the forcing is nontrivial and the domain is connected, then

$$h(y) > 0 \quad \text{in the interior.} \quad (3.4)$$

This yields the geometric safe set

$$\mathcal{C} = \{p : h(p) \geq 0\}. \quad (3.5)$$

#### Key idea

Obstacles are encoded through the domain  $\Omega$  and the boundary condition  $h = 0$  on  $\partial\Omega$ . The forcing field is not an obstacle mask; it changes the curvature and gradient distribution within free space.

## 3.2 Variational formulation

With  $q = -f_P$ , the weak formulation seeks  $h \in H_0^1(\Omega)$  such that

$$\int_{\Omega} \nabla h^\top \nabla \varphi \, dy = \int_{\Omega} q \varphi \, dy \quad \forall \varphi \in H_0^1(\Omega). \quad (3.6)$$

Equivalently,  $h$  minimizes

$$\mathcal{E}(w) = \frac{1}{2} \int_{\Omega} \|\nabla w\|^2 \, dy - \int_{\Omega} q w \, dy. \quad (3.7)$$

This interpretation clarifies the field's smoothness. The first term penalizes rapid spatial variation, while the load term pushes the interior toward positive values.

### 3.3 Four forcing-field families

#### 3.3.1 Constant forcing

The library uses

$$f_P(y) = -c, \quad c > 0. \quad (3.8)$$

This is the simplest reference case and isolates the effect of domain geometry.

#### 3.3.2 Distance-based forcing

Let  $d(y)$  be the distance from a solve cell to the complement of the solve mask and  $d_{\max} = \|d\|_{\infty}$ . The 2-D implementation uses

$$f_P(y) = -\left(\frac{d(y)}{d_{\max}}\right)^{\alpha}, \quad \alpha > 0. \quad (3.9)$$

The supplied static experiment uses  $\alpha = 0.55$ .

#### 3.3.3 Average-flux forcing

The library approximates boundary measure and free-space area or volume, and sets

$$f_P = \bar{b} \frac{|\partial\Omega|}{|\Omega|}, \quad \bar{b} < 0. \quad (3.10)$$

The motivation follows from the divergence theorem:

$$\int_{\Omega} \Delta h \, dy = \int_{\partial\Omega} \nabla h^{\top} n \, dS. \quad (3.11)$$

Thus, the integral of the forcing determines the total normal gradient flux through the boundary.

#### 3.3.4 Guidance forcing

The library first estimates a boundary normal using a signed-distance field and defines

$$v_{\partial}(y) = b(y)n(y). \quad (3.12)$$

Each vector component is harmonically extended:

$$\Delta v_j = 0 \quad \text{in } \Omega, \quad v_j = (v_{\partial})_j \quad \text{on } \partial\Omega. \quad (3.13)$$

The divergence is then passed through a negative softplus:

$$f_P(y) = -\frac{1}{\beta} \log(1 + \exp[-\beta \nabla \cdot v(y)]). \quad (3.14)$$

Finally, the mean magnitude is rescaled. This procedure shapes preferred gradient directions without introducing hard discontinuities.

### Implementation note

The standalone 3-D contingency script uses the same four forcing families, but with its own numerical parameterizations. They are not bit-for-bit identical to the functions in `poisson_safety_box`. In particular, its average-flux variant is scaled by mean clearance, and its guidance field combines target direction with clearance normals.

## 3.4 Finite-difference discretization

### 3.4.1 Two-dimensional stencil

On a Cartesian grid,

$$\Delta h_{i,j} \approx \frac{h_{i+1,j} - 2h_{i,j} + h_{i-1,j}}{\Delta x^2} + \frac{h_{i,j+1} - 2h_{i,j} + h_{i,j-1}}{\Delta y^2}. \quad (3.15)$$

The implementation assembles  $-\Delta h = -f_P$ , so each unknown produces

$$A_{kk} = 2 \left( \Delta x^{-2} + \Delta y^{-2} \right), \quad (3.16)$$

$$A_{k,k\pm x} = -\Delta x^{-2}, \quad (3.17)$$

$$A_{k,k\pm y} = -\Delta y^{-2}. \quad (3.18)$$

### 3.4.2 Three-dimensional stencil

Two neighbors in the  $z$  direction are added:

$$A_{kk} = 2 \left( \Delta x^{-2} + \Delta y^{-2} + \Delta z^{-2} \right), \quad (3.19)$$

$$A_{k,k\pm z} = -\Delta z^{-2}. \quad (3.20)$$

Boundary cells are not unknowns. Their prescribed values are moved to the right-hand side. In the supplied examples, the Dirichlet value is zero, so those contributions vanish.

## 3.5 Sparse linear system and solvers

The result is the sparse system

$$\boxed{A\mathbf{h} = \mathbf{b}.} \quad (3.21)$$

For a well-defined Dirichlet problem,  $A$  is sparse, symmetric, and positive definite. The implementations compare:

- **sparse direct:** direct factorization, high numerical accuracy, and higher memory cost;
- **conjugate gradient:** an iterative method for SPD matrices;
- **SOR:** successive over-relaxation, simple to implement, although a step-change stopping criterion may not control the true algebraic residual;
- **BiCGSTAB:** used in the 3-D study as a general iterative alternative.

Solver validation must not be limited to a success flag. The algebraic residual is reconstructed as

$$r = A\mathbf{h} - \mathbf{b}, \quad (3.22)$$

and the following quantities are reported:

$$\|r\|_2, \quad \|r\|_\infty, \quad \frac{\|r\|_2}{\|\mathbf{b}\|_2}. \quad (3.23)$$

The boundary consistency check is

$$\max_{q \in \partial\Omega_h} |h(q)|. \quad (3.24)$$

### 3.6 Gradient, Hessian, and interpolation

After solving for  $h$ , the implementation computes

$$\nabla h = \begin{bmatrix} h_x & h_y & h_z \end{bmatrix}^\top, \quad \nabla^2 h = \begin{bmatrix} h_{xx} & h_{xy} & h_{xz} \\ h_{yx} & h_{yy} & h_{yz} \\ h_{zx} & h_{zy} & h_{zz} \end{bmatrix}. \quad (3.25)$$

The code uses finite differences through `numpy.gradient`; the Hessian is symmetrized to reduce numerical asymmetry.

The robot almost never lies exactly on a grid node. In 2-D, if  $p$  lies inside a cell with fractional coordinates  $\xi, \eta \in [0, 1]$ , bilinear interpolation gives

$$h(p) = (1 - \xi)(1 - \eta)h_{00} + \xi(1 - \eta)h_{10} \quad (3.26)$$

$$+ (1 - \xi)\eta h_{01} + \xi\eta h_{11}. \quad (3.27)$$

The same weights are applied to gradients and Hessians. Trilinear interpolation uses eight neighbors in 3-D.

#### 3.6.1 Axis conversion

Image arrays are indexed as (row, column) =  $(y, x)$ , while the controller uses  $(x, y)$ . Therefore,

$$\nabla_{xy} h = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \nabla_{yx} h, \quad (3.28)$$

and

$$H_{xy} = P H_{yx} P^\top, \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (3.29)$$

An incorrect transpose changes the orientation of the CBF half-space and can turn a safe correction into an unsafe one.

# Control Barrier Functions and QP Safety Filters

## 4.1 Safe set and forward invariance

Let

$$\mathcal{C} = \{x : h(x) \geq 0\}. \quad (4.1)$$

For a control-affine system

$$\dot{x} = f(x) + g(x)u, \quad (4.2)$$

a relative-degree-one CBF condition is

$$L_f h(x) + L_g h(x)u \geq -\alpha(h(x)), \quad (4.3)$$

where  $\alpha$  is an extended class- $\mathcal{K}$  function. If a locally Lipschitz controller satisfies this inequality and the initial state belongs to  $\mathcal{C}$ , then, under the standard CBF assumptions,  $\mathcal{C}$  is forward invariant.

### Technical warning

The theorem is continuous-time and model dependent. Sampling, latency, saturation, state-estimation error, interpolation error, and unmodeled dynamics can violate its assumptions. For this reason, the first example adds a discrete backtracking guard, while the second includes acceleration limits directly in the QP.

## 4.2 Velocity-level CBF for the single integrator

For

$$\dot{p} = u, \quad (4.4)$$

the safety derivative is

$$\dot{h} = \nabla h(p)^\top u. \quad (4.5)$$

Using  $\alpha(h) = \gamma h$  gives

$$\boxed{\nabla h(p)^\top u \geq -\gamma h(p).} \quad (4.6)$$

The safety filter minimizes deviation from the nominal command:

$$\begin{aligned} u_{\text{safe}} &= \arg \min_u \frac{1}{2} \|u - u_{\text{nom}}\|^2 \\ \text{subject to} \quad &\nabla h^\top u \geq -\gamma h. \end{aligned} \quad (4.7)$$

### 4.2.1 Closed-form solution for one inequality

Let  $g = \nabla h$  and  $b = -\gamma h$ . If  $g^\top u_{\text{nom}} \geq b$ , then  $u_{\text{safe}} = u_{\text{nom}}$ . Otherwise,

$$u_{\text{safe}} = u_{\text{nom}} + \frac{b - g^\top u_{\text{nom}}}{\|g\|^2} g. \quad (4.8)$$

This is the Euclidean projection of the nominal velocity onto the safe half-space.

## 4.3 Discrete integration and backtracking

Even if (4.6) is satisfied at  $p_k$ , a large Euler step

$$p_{k+1} = p_k + \Delta t u_{\text{safe}} \quad (4.9)$$

can jump into an occupied cell. The first implementation tests the candidate state and, when it is invalid, repeatedly applies

$$\Delta t \leftarrow \frac{1}{2} \Delta t \quad (4.10)$$

until the new point is free, interpolable, and satisfies  $h \geq -\varepsilon_h$ . This guard does not replace the CBF; it protects the numerical integration layer.

## 4.4 Acceleration-level HOCBF for the double integrator

For

$$\dot{p} = v, \quad \dot{v} = a, \quad (4.11)$$

the first derivative is

$$\dot{h} = \nabla h^\top v, \quad (4.12)$$

and the second derivative is

$$\ddot{h} = \nabla h^\top a + v^\top \nabla^2 h v. \quad (4.13)$$

Define

$$\psi_0 = h, \quad \psi_1 = \dot{h} + \alpha_1 h. \quad (4.14)$$

Enforcing

$$\dot{\psi}_1 + \alpha_2 \psi_1 \geq 0 \quad (4.15)$$

produces

$$\nabla h^\top a + v^\top \nabla^2 h v + (\alpha_1 + \alpha_2) \nabla h^\top v + \alpha_1 \alpha_2 h \geq 0. \quad (4.16)$$

As an affine QP row,

$$\nabla h^\top a \geq -v^\top \nabla^2 h v - (\alpha_1 + \alpha_2) \nabla h^\top v - \alpha_1 \alpha_2 h. \quad (4.17)$$

### Key idea

The curvature term  $v^\top \nabla^2 h v$  is not optional. It describes how the safety gradient changes as the vehicle moves. Omitting it gives an incorrect second derivative of the barrier.

## 4.5 CLF for the active task

For a reference position  $p_r$ , define the double-integrator error

$$e = \begin{bmatrix} p - p_r \\ v \end{bmatrix}. \quad (4.18)$$

A nominal PD controller has the form

$$a = -K_p e_p - K_d v. \quad (4.19)$$

A quadratic CLF is

$$V = e^\top P e, \quad P \succ 0, \quad (4.20)$$

where  $P$  satisfies

$$A_{cl}^\top P + P A_{cl} = -Q, \quad Q \succ 0. \quad (4.21)$$

The CLF inequality is

$$\dot{V} = \nabla_p V^\top v + \nabla_v V^\top a \leq -\gamma_V V. \quad (4.22)$$

Unlike collision avoidance, the task objective may be relaxed when it conflicts with hard safety constraints.

# Multi-Zone Contingency

## 5.1 Smooth certificate for each landing zone

For landing zone  $L_i$  with center  $\ell_i$ , define

$$e_i = \begin{bmatrix} p - \ell_i \\ v \end{bmatrix}. \quad (5.1)$$

The prototype uses

$$W_i(p, v) = \frac{e_i^\top P e_i}{s_i} + w_g \left( \frac{d_i^{\text{geo}}(p_{xy})}{D_{xy}} \right)^2 + w_h \phi(h(p))^2 + w_v \frac{\|v\|^2}{v_{\max}^2}. \quad (5.2)$$

The four terms represent, respectively:

- dynamic error relative to the landing zone;
- obstacle-aware geodesic travel effort;
- risk associated with low Poisson-field values;
- kinetic or braking effort.

The smooth Poisson-risk penalty is

$$\phi(h) = \tau \log \left( 1 + \exp \left( \frac{h_{\min} - h}{\tau} \right) \right). \quad (5.3)$$

## 5.2 Budget and viability margin

The zone budget shrinks with time, energy use, and blockage:

$$c_i(t, E) = c_{i0} - \beta_t t - \beta_E E(t) - B_i(t). \quad (5.4)$$

The certificate is

$$\rho_i = c_i - W_i. \quad (5.5)$$

Within the prototype,

$$\rho_i \geq 0 \quad (5.6)$$

means that the estimated cost of reaching zone  $i$  fits within its allocated budget.

### 5.2.1 Derivative affine in acceleration

Since

$$\dot{W}_i = \nabla_p W_i^\top v + \nabla_v W_i^\top a, \quad (5.7)$$

we obtain

$$\dot{\rho}_i = \dot{c}_i - \nabla_p W_i^\top v - \nabla_v W_i^\top a. \quad (5.8)$$

Acceleration appears linearly, which makes the certificate compatible with a QP.

### 5.3 The $r$ -out-of- $p$ pivot

Sort the certificates in descending order:

$$\rho_{(1)} \geq \rho_{(2)} \geq \dots \geq \rho_{(p)}. \quad (5.9)$$

Define

$$\tilde{\rho} = \rho_{(r)} = \max_i^{(r)} \rho_i. \quad (5.10)$$

Then

$$\tilde{\rho} \geq 0 \iff \text{at least } r \text{ certificates are nonnegative.} \quad (5.11)$$

### 5.4 Smooth combinatorial constraints

Directly differentiating an order statistic introduces nonsmooth switching when the pivot changes. The implemented construction instead uses  $p$  smooth rows:

$$\dot{\rho}_i \geq -\gamma_c \rho_i - \omega R(\rho_i - \tilde{\rho}), \quad i = 1, \dots, p, \quad (5.12)$$

with

$$R(s) = k_R s^2, \quad \omega \geq 0. \quad (5.13)$$

For a pivot-critical certificate,  $\rho_i = \tilde{\rho}$  and  $R(0) = 0$ , so the auxiliary variable cannot relax the critical row. For noncritical certificates, the auxiliary term reduces incompatibility without requiring every zone to remain viable.

Substituting (5.8) gives

$$-\nabla_v W_i^\top a + R_i \omega \geq -\gamma_c \rho_i - \dot{c}_i + \nabla_p W_i^\top v, \quad (5.14)$$

where

$$R_i = k_R (\rho_i - \tilde{\rho})^2. \quad (5.15)$$

### 5.5 Complete QP

The decision vector is

$$z = \begin{bmatrix} a \\ \omega \end{bmatrix} \in \mathbb{R}^4. \quad (5.16)$$

The optimization problem is

$$\begin{array}{ll} \min_{a, \omega} & \frac{1}{2} \|a - a_{\text{nom}}\|^2 + c_\omega \omega^2 \\ \text{subject to} & \text{Poisson environmental HOCBF,} \\ & \text{active landing-zone CLF,} \\ & p \text{ contingency inequalities,} \\ & \|a\|_2 \leq a_{\text{max}}, \quad \omega \in [0, \omega_{\text{max}}]. \end{array} \quad (5.17)$$

The primary implementation has four decision variables. Its custom solver performs a dual active-set projection, adds tangent cuts for the acceleration ball, and includes a heavily penalized shared emergency slack. If infeasibility is caused only by the task row, the code resolves the problem without the active CLF before compromising the hard environmental or contingency rows.

## 5.6 Safe target switching

When a landing zone is declared blocked, its budget collapses smoothly and it is subsequently marked unavailable. Among valid alternatives, the code evaluates

$$S_i^{\text{switch}} = 2.25\rho_i + 0.34 s_i^{\text{science}} + 0.16 q_i^{\text{terrain}} - 0.028 d_i^{\text{geo}}. \quad (5.18)$$

The highest-scoring alternative is selected. The environmental HOCBF remains active before, during, and after the target switch.

### Scope of the claim

A formal Hamilton–Jacobi implementation would replace the proxies with value functions  $V_i^{HJ}(p, v, t)$  and could define  $\rho_i = -V_i^{HJ}$ . The cost would be solving a reach–avoid PDE in a high-dimensional state space. The present prototype avoids that cost, but it also gives up the exact backward–reach–avoid interpretation.

# Software Architecture

## 6.1 Separation into modular boxes

The supplied implementation is organized into independent modules:

- `poisson_safety_box`: receives occupancy and returns  $h$ ,  $\nabla h$ ,  $\nabla^2 h$ , masks, and numerical diagnostics;
- `cbf_safety_box`: receives a local safety sample and a nominal command, constructs constraints, and solves a safety-filter QP;
- `vision_poisson_experiments`: performs calibration, segmentation, inflation, coordinate conversion, experiment execution, and data export;
- the contingency study script: constructs the 3-D world, multi-zone proxies, augmented QP, failure event, and parameter sweeps.

## 6.2 Data contract between Poisson and CBF

The local interface is essentially

$$\text{SafetySample} = \{h(p), \nabla h(p), \nabla^2 h(p), \Delta h(p), \text{metadata}\}. \quad (6.1)$$

The CBF layer does not need to know the original image, the analytic obstacle list, or which linear solver produced  $h$ . This decoupling allows perception, PDE synthesis, and control to be replaced independently.

## 6.3 Pseudocode for the Poisson pipeline

**Listing 6.1:** Simplified structure of the Poisson safety box.

```
def compute_poisson(occupancy, config):
    occupied = validate_boolean(occupancy)
    free = ~occupied
    boundary = free_cells_touching_occupied(occupied)
    boundary |= optional_outer_wall(free)
    solve_mask = free & ~boundary

    f = build_forcing(config.method, solve_mask, boundary)
    A, b = assemble_minus_laplacian(f, solve_mask, h_boundary=0)
    h_unknown = solve(A, b, config.solver)
    h = scatter(h_unknown, boundary_value=0)

    grad_h = finite_difference_gradient(h)
    hessian_h = finite_difference_hessian(h)
    validate(A @ h_unknown - b, h[boundary])
    return h, grad_h, hessian_h
```

## 6.4 Pseudocode for the static velocity filter

**Listing 6.2:** Velocity-level CBF used in Case Study 1.

```

while not at_goal:
    h, grad_h = bilinear_sample(field, position)
    u_nom = saturate(k_goal * (goal - position), u_max)
    u_safe = project_to_halfspace(
        u_nom,
        normal=grad_h,
        lower_bound=-alpha * h,
    )
    assert grad_h @ u_safe + alpha * h >= -tolerance

    dt_trial = dt
    while candidate_is_not_discretely_safe(position + dt_trial*u_safe):
        dt_trial *= 0.5
    position += dt_trial * u_safe

```

## 6.5 Pseudocode for the contingency-aware filter

**Listing 6.3:** Main loop of Case Study 2.

```

for each sample time t:
    sample h, grad_h, Hessian_h at (p, v)
    evaluate every W_i, c_i, rho_i
    pivot = rth_largest(available rho_i, r)

    if active site is blocked or loses margin:
        active = select_best_available_backup()

    a_nom = Kp * (reference(active) - p) - Kd * v
    build environment HOCBF row
    build active-target CLF row
    build p combinatorial rows
    solve min 0.5*||a-a_nom||^2 + c_omega*omega^2
    integrate p_dot=v, v_dot=a

```

## Part II

### Case Study 1: Static Image, 2-D Poisson Field, and Velocity CBF

# Configuration and Objective of Case Study 1

## 7.1 Experimental question

This experiment asks whether a single overhead image can be converted into a numerical field that allows a single-integrator robot to reach a goal while avoiding detected objects. Two controllers are compared:

- a saturated nominal velocity that points directly toward the goal;
- the same nominal velocity after projection through the Poisson CBF.

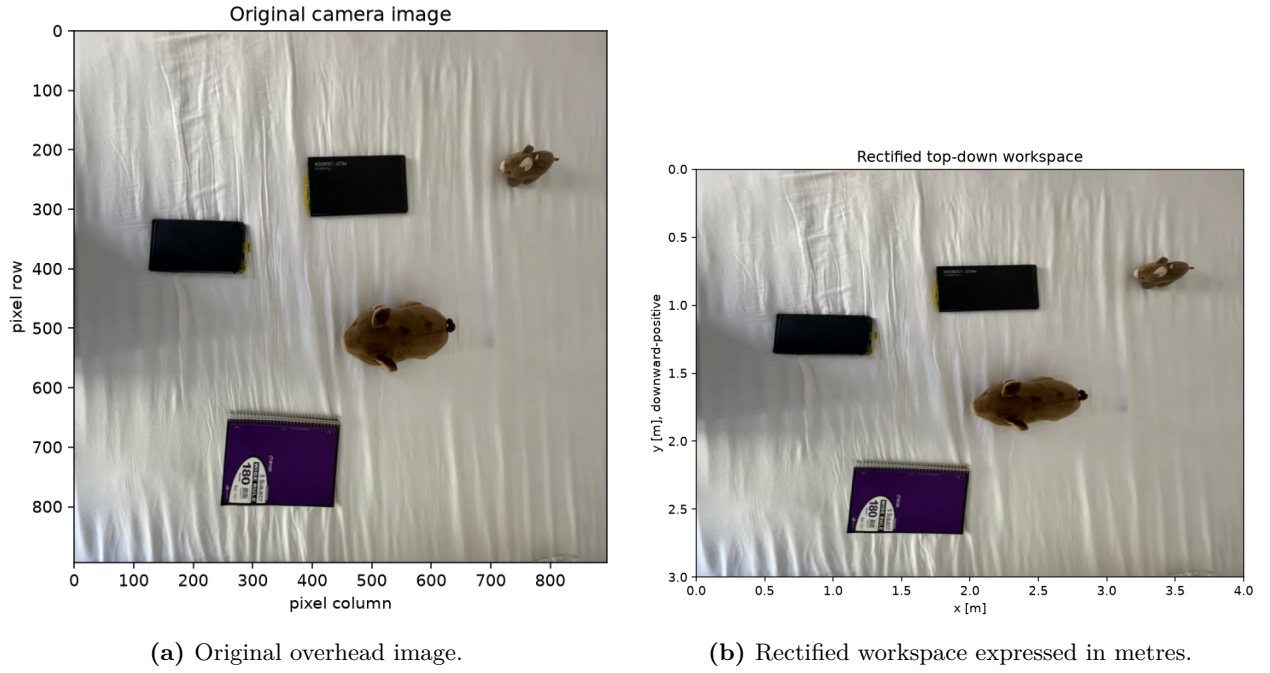
## 7.2 Configuration of the supplied run

**Table 7.1:** Primary configuration of Case Study 1.

| Quantity           | Value            | Quantity          | Value            |
|--------------------|------------------|-------------------|------------------|
| Physical workspace | $4 \times 3$ m   | Grid              | $80 \times 60$   |
| Start              | $(0.35, 1.50)$ m | Goal              | $(3.65, 1.50)$ m |
| Robot radius       | 0.08 m           | Perception margin | 0.04 m           |
| Total inflation    | 0.12 m           | Primary forcing   | constant         |
| Primary solver     | sparse direct    | CBF gain $\alpha$ | 3.0              |
| Maximum speed      | 0.5 m/s          | Nominal step      | 0.02 s           |

# Visual Procedure, Step by Step

## 8.1 Step 1: original image and metric rectification



**Figure 8.1:** The homography transforms the camera image into a metric rectangle. In this supplied run, an explicit top-down-view assumption makes the transformation close to a rescaling, but the calibration stage is retained so the same pipeline can support perspective correction.

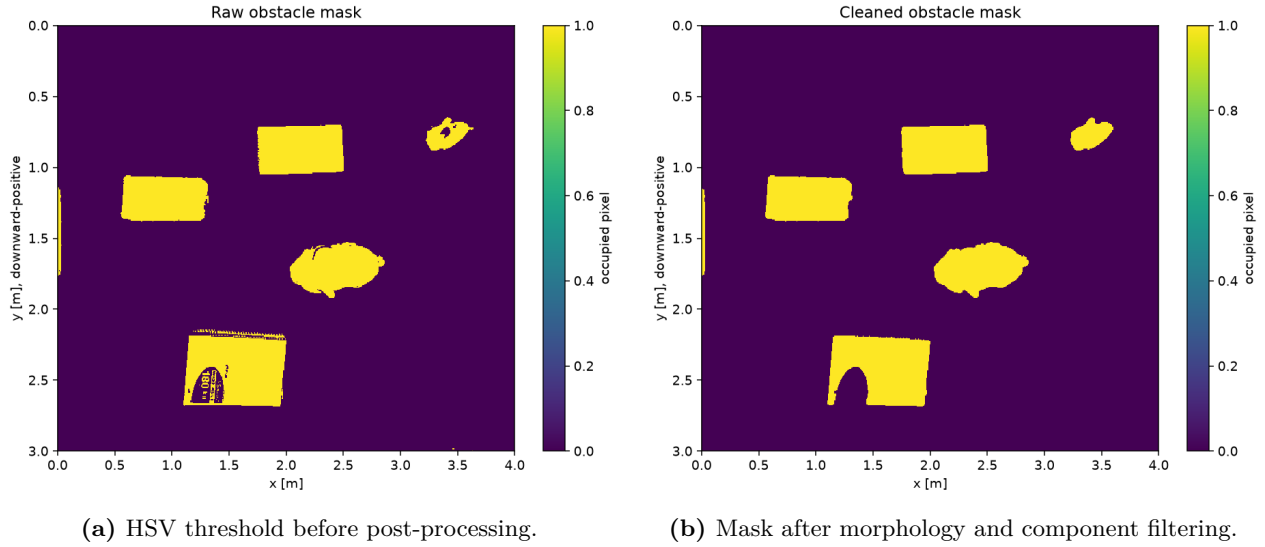
The image contains five dark objects over a light background. Rectification maps the horizontal image direction to physical  $x$  and uses a positive  $y$  direction downward in the displayed map. This convention matches the image array and avoids a hidden reflection, but it must be remembered when interpreting vertical control components.

## 8.2 Step 2: raw and cleaned perception masks

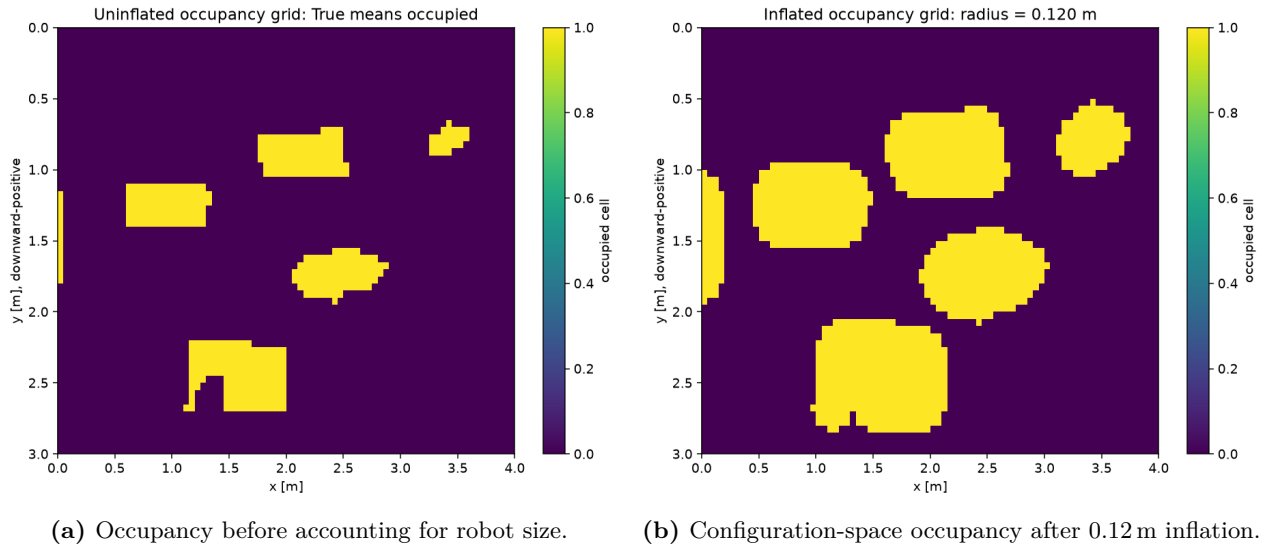
The supplied thresholds are  $H \in [0, 179]$ ,  $S \in [0, 255]$ , and  $V \in [0, 105]$ . Consequently, the detector selects dark regions largely independently of hue and saturation. The cleaned result occupies 445 cells of the Poisson grid before geometric inflation.

## 8.3 Step 3: occupancy and configuration-space inflation

With  $\Delta x \approx 0.0506$  m and  $\Delta y \approx 0.0508$  m, the implementation uses an inflation radius of three cells along each axis. The occupied fraction increases from approximately 9.27% to 24.65%.



**Figure 8.2:** Segmentation does not directly produce the Poisson domain. It first generates a perception mask, which is cleaned so that small noise is removed and the retained obstacle components are spatially coherent.



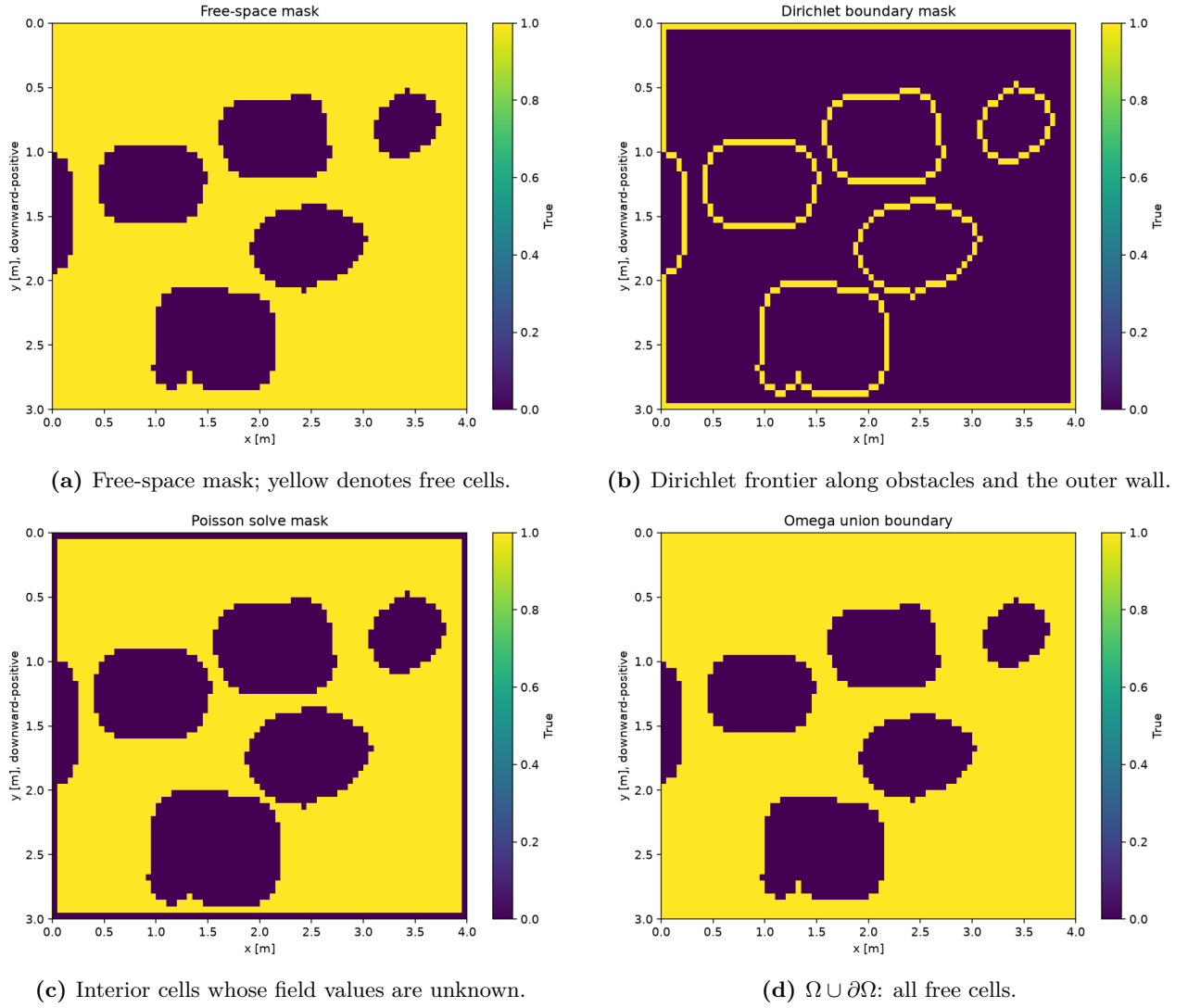
**Figure 8.3:** Inflation increases the number of occupied cells from 445 to 1183. Every later safety calculation is evaluated against the inflated occupancy, not against the original visible silhouette.

## 8.4 Step 4: construction of $\Omega$ , $\partial\Omega$

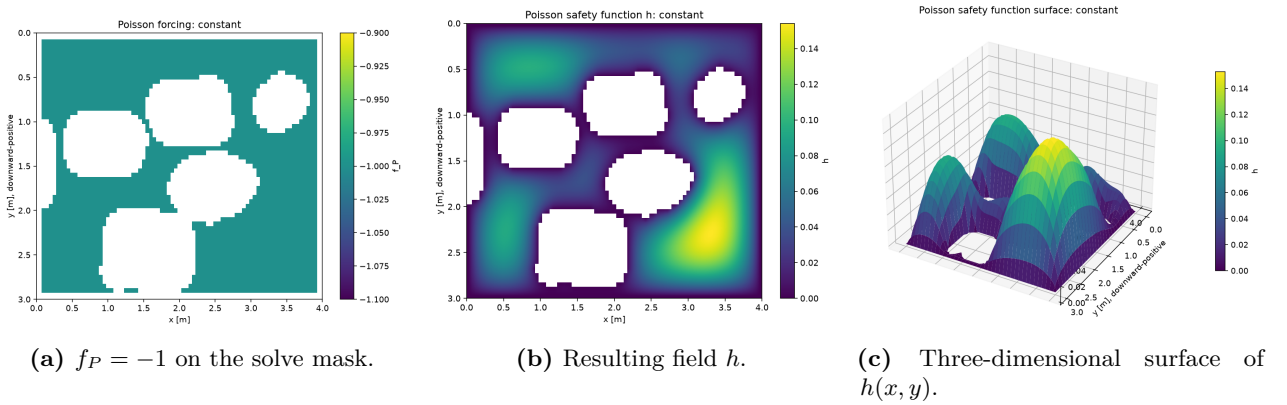
This step is conceptually important because it separates three objects that are often confused. Occupancy marks points that are not part of the free domain. Boundary cells are free points whose safety value is prescribed. Solve cells are free interior points whose values are determined by the Poisson equation.

## 8.5 Step 5: forcing and Poisson solution

For constant forcing in the supplied run, the maximum interior value is approximately 0.1541, and the mean over solve cells is approximately 0.0518. The maximum algebraic residual of the sparse direct solve is approximately  $1.56 \times 10^{-13}$ , while the prescribed boundary remains zero to numerical precision.

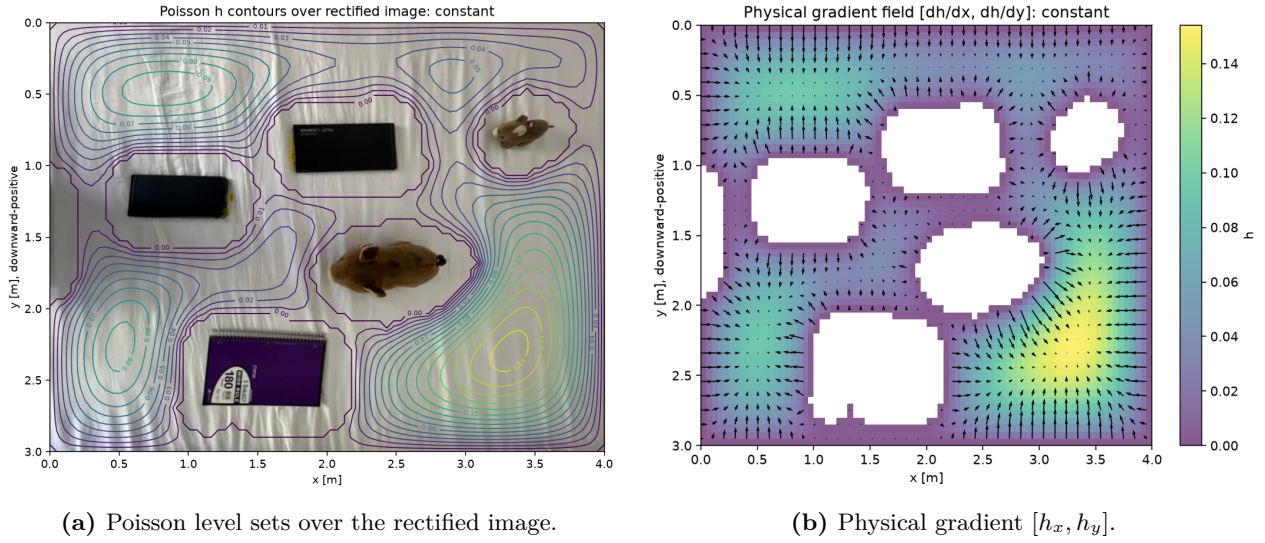


**Figure 8.4:** Discrete decomposition of the computational domain. The PDE is solved at 3065 interior cells, while 552 free cells form the Dirichlet frontier.



**Figure 8.5:** The field is zero on Dirichlet boundaries and forms positive lobes in broad free regions. It is not Euclidean distance: its magnitude depends on domain geometry, forcing, and spatial scaling.

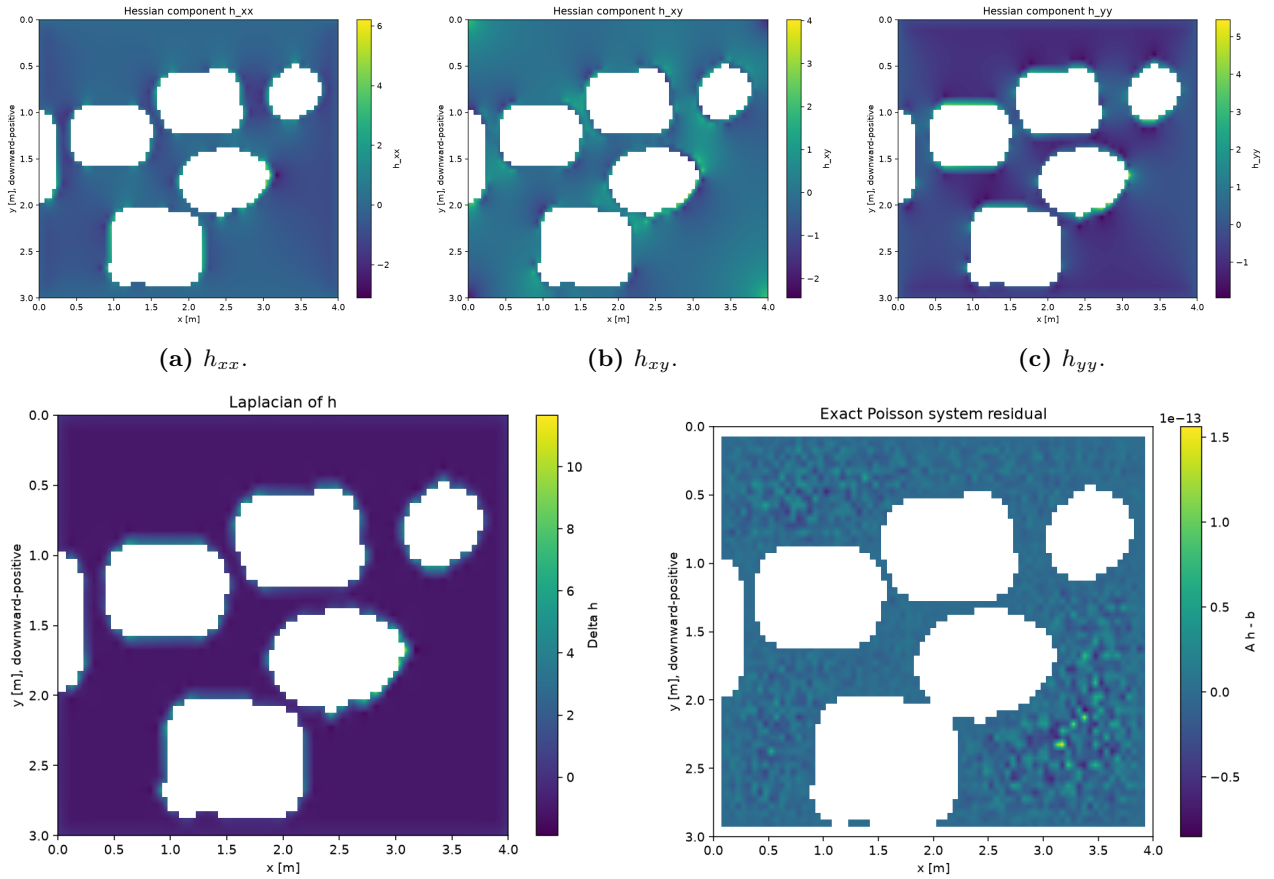
## 8.6 Step 6: level sets and the safety gradient



**Figure 8.6:** All obstacles jointly deform the level sets. The gradient is normal to those level sets and therefore determines the orientation of the safe velocity half-space.

Near a boundary, a command pointing toward an obstacle produces  $\nabla h^\top u < 0$ . The QP corrects the normal component responsible for decreasing safety. A tangential component can remain nearly unchanged when it does not violate the CBF inequality.

## 8.7 Step 7: Hessian, Laplacian, and residual verification



(d) Laplacian reconstructed from numerical derivatives.

(e) Exact residual of the assembled sparse system.

**Figure 8.7:** Second derivatives concentrate near boundaries, where field curvature changes rapidly. The exact residual  $A\mathbf{h} - \mathbf{b}$  is the primary solver verification. The Laplacian reconstructed with `numpy.gradient` is a secondary diagnostic because it uses a different numerical stencil.

# Numerical Comparisons in Case Study 1

## 9.1 Comparison of forcing fields

All four fields use exactly the same occupancy and the same Dirichlet frontier. Therefore, every interior difference is caused by the forcing, not by a change in obstacle geometry. Numerical validation precedes visual interpretation:

**Table 9.1:** Validation of the four forcing fields in the supplied static experiment.

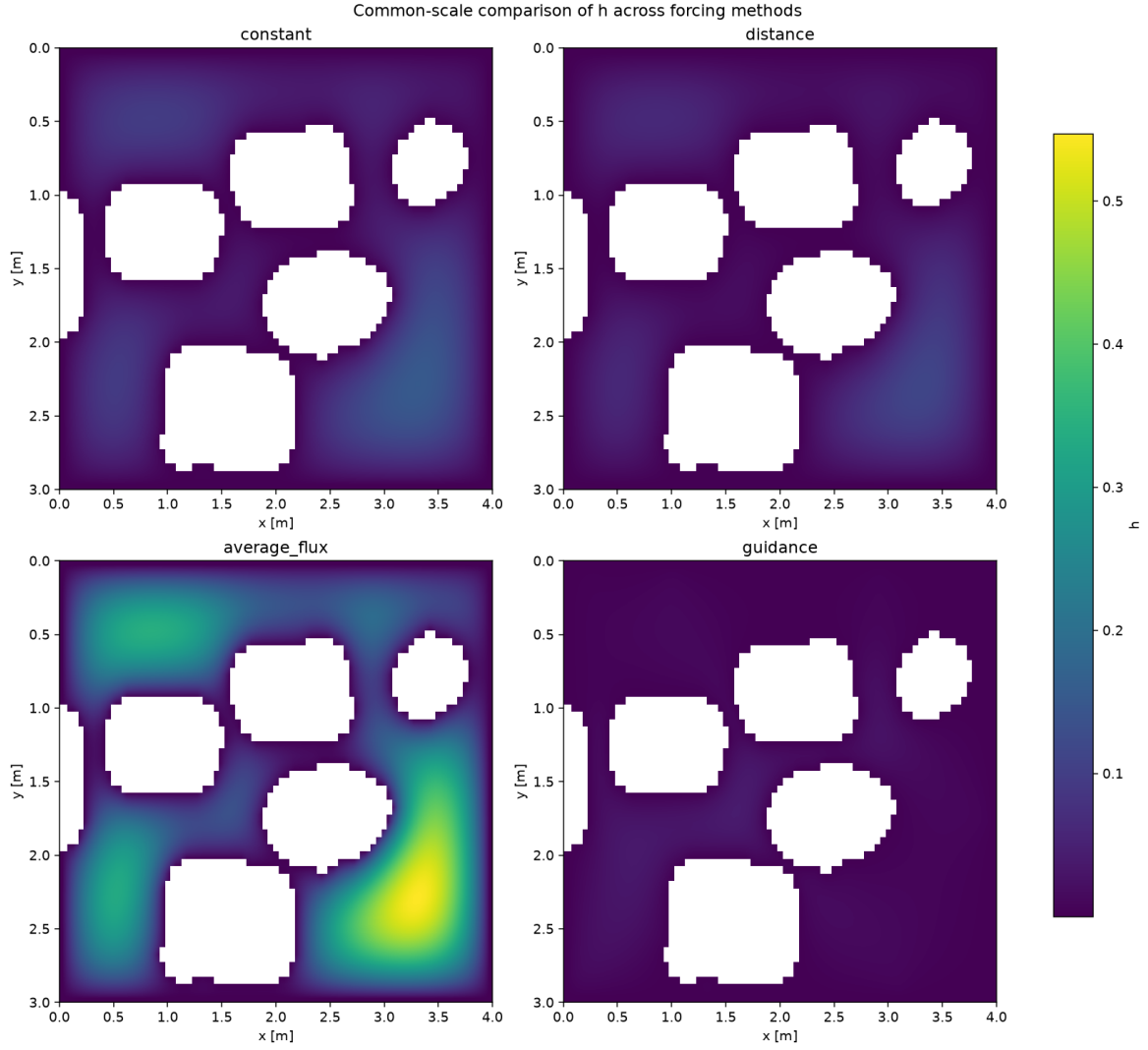
| Method       | Time [s] | $\ A\mathbf{h} - \mathbf{b}\ _\infty$ | $h_{\max}$ |
|--------------|----------|---------------------------------------|------------|
| constant     | 0.020 7  | $1.56 \times 10^{-13}$                | 0.154 1    |
| distance     | 0.020 7  | $8.56 \times 10^{-14}$                | 0.115 2    |
| average flux | 0.020 2  | $3.21 \times 10^{-13}$                | 0.547 0    |
| guidance     | 0.061 9  | $2.40 \times 10^{-14}$                | 0.041 2    |

The value  $h_{\max}$  is not a universal measure of field quality. A larger amplitude may simply reflect a larger forcing scale. For control, the important properties are the local geometry of  $h$  and  $\nabla h$ , numerical consistency, derivative regularity, and QP feasibility.

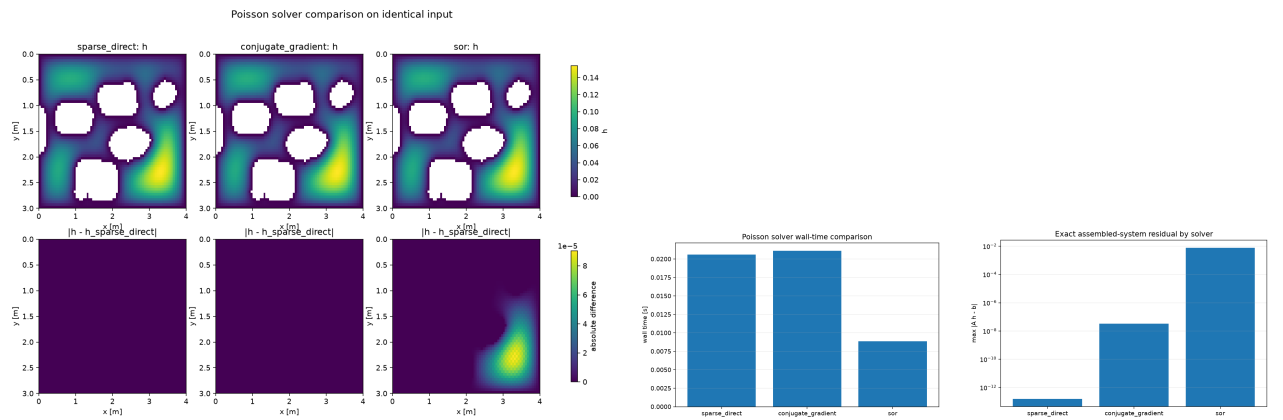
## 9.2 Comparison of Poisson solvers

### Observed result

Sparse direct and conjugate gradient were valid. SOR reported convergence using a maximum-update criterion, but its exact maximum residual was approximately  $7.89 \times 10^{-3}$ , above the requested tolerance of  $10^{-5}$ . The experiment therefore marks that result as invalid. This is a concrete example of why an internal stopping rule and independent PDE verification must be treated separately.



**Figure 9.1:** Comparison of  $h$  under a common color scale for constant, distance, average-flux, and guidance forcing. The geometric boundary is identical in all panels; only the interior shape and amplitude change.



(a) Fields produced from the same Poisson problem.

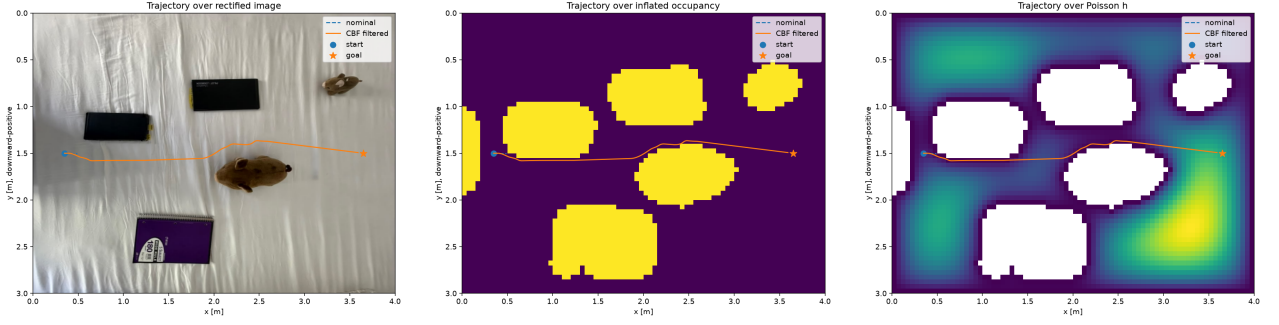
(b) Wall-clock time.

(c) Exact algebraic residual.

**Figure 9.2:** Sparse direct, conjugate-gradient, and SOR comparison. A solver's internal convergence flag does not replace verification of  $Ah - b$ .

# CBF Result of Case Study 1

## 10.1 Nominal trajectory versus filtered trajectory



(a) Trajectories over the image.

(b) Trajectories over inflated occupancy.

(c) Trajectories over the Poisson field.

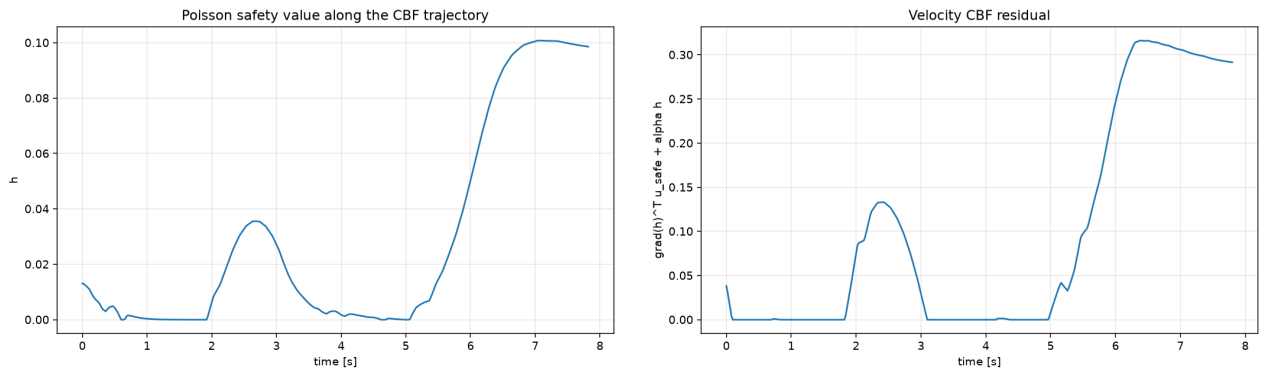
**Figure 10.1:** The straight nominal trajectory intersects inflated occupancy. The Poisson-CBF trajectory creates vertical components, passes around the central obstacle, and reaches the goal.

The nominal controller is

$$u_{\text{nom}} = \text{sat}_{0.5}(1.25(p_g - p)). \quad (10.1)$$

The filter uses (4.7) with  $\alpha = 3$ . When the nominal velocity already belongs to the safe half-space, it is not modified. When it points too strongly toward a boundary, it is projected onto the active CBF plane.

## 10.2 Evolution of safety

(a) Sampled  $h$  along the safe trajectory.(b) Residual  $\nabla h^T u_{\text{safe}} + \alpha h$ .

**Figure 10.2:** The explicit CBF residual remains nonnegative up to floating-point roundoff. The field approaches zero where the trajectory runs close to the numerical safe-set frontier.

The minimum recorded residual is approximately  $-1.04 \times 10^{-17}$ , which is consistent with floating-point error. The run summary records  $\min h = 0$ , no transition into occupied cells, and successful arrival at the goal.

Figure 10 consists of four subplots arranged in a 2x2 grid, labeled (a) through (d).

- (a) Nominal and safe  $x$  components.** The plot shows  $x$  velocity [m/s] on the y-axis (0.0 to 0.5) versus time [s] on the x-axis (0 to 8). Two series are shown:  $u_{\text{nom}_x}$  (blue line) and  $u_{\text{safe}_x}$  (orange line).  $u_{\text{nom}_x}$  is a constant 0.5 m/s until approximately 6.2 seconds, then decreases to 0.0 m/s at 8 seconds.  $u_{\text{safe}_x}$  follows  $u_{\text{nom}_x}$  but has several sharp drops to approximately 0.3 m/s at 0.5s, 3.5s, 4.5s, and 5.5s, and a final drop to 0.0 m/s at 8 seconds.
- (b) Nominal and safe  $y$  components.** The plot shows  $y$  velocity [m/s] on the y-axis (-0.2 to 0.2) versus time [s] on the x-axis (0 to 8). Two series are shown:  $u_{\text{nom}_y}$  (blue line) and  $u_{\text{safe}_y}$  (orange line).  $u_{\text{nom}_y}$  starts at 0.0 m/s, dips slightly, then rises to approximately 0.05 m/s at 4.5 seconds, and remains constant until 8 seconds.  $u_{\text{safe}_y}$  shows significant fluctuations, with peaks around 0.2 m/s and troughs around -0.2 m/s, before settling near 0.0 m/s after 6.5 seconds.
- (c) CBF intervention magnitude.** The plot shows  $\text{norm}(u_{\text{safe}} - u_{\text{nom}})$  [m/s] on the y-axis (0.00 to 0.35) versus time [s] on the x-axis (0 to 8). The blue line shows several sharp peaks, with the highest peak reaching approximately 0.34 m/s at 4.5 seconds. The magnitude is zero for most of the time interval.
- (d) Geometric clearance to inflated occupancy (diagnostic only).** The plot shows clearance [m] on the y-axis (0.0 to 0.5) versus time [s] on the x-axis (0 to 8). The blue line shows a peak of approximately 0.3 m at 2.8 seconds, a dip, and then a steady increase to approximately 0.5 m by 8 seconds.

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## 10.4 Conclusion and unresolved diagnostics

### Observed result

- Nominal control: collision with inflated occupancy.
- Poisson-CBF control: goal reached without a discrete collision.
- Minimum explicit CBF residual: approximately  $-10^{-17}$ .
- The primary Poisson solve satisfies the assembled system to approximately  $10^{-13}$  in the infinity norm.

### Technical warning

The directional-derivative verification file reports an absolute difference of approximately  $2.48 \times 10^{-2}$  between a centered finite-difference directional derivative and the interpolated gradient at the evaluated point. This deserves further study. It may reflect step-size selection, proximity to high curvature, mixed numerical stencils, or a coordinate-conversion issue. It does not invalidate the algebraic Poisson residual, but it does show that the complete chain *discrete derivative*  $\rightarrow$  *interpolation*  $\rightarrow$  *controller* should be verified systematically before hardware deployment.

## Part III

### **Case Study 2: 3-D Landing with Poisson-HOCBF and Contingency**

# Configuration of Case Study 2

## 11.1 World, landing zones, and dynamics

The simulated world occupies

$$[0, 18] \times [0, 14] \times [0, 10] \text{ m} \quad (11.1)$$

and is discretized with a  $48 \times 38 \times 28$  grid. The initial condition is

$$p_0 = (1.0, 1.1, 8.6) \text{ m}, \quad v_0 = 0. \quad (11.2)$$

The reduced-order model is

$$\dot{p} = v, \quad \dot{v} = a, \quad \|a\|_2 \leq 1.35 \text{ m/s}^2, \quad \|v\|_2 \leq 2.0 \text{ m/s}. \quad (11.3)$$

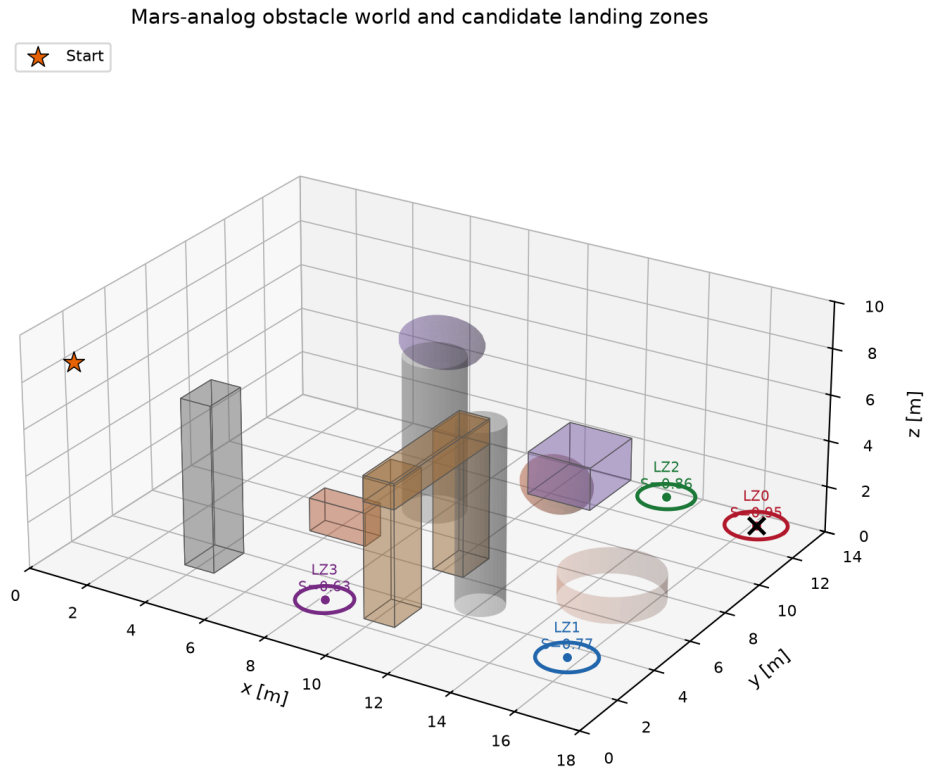
**Table 11.1:** Candidate landing zones in Case Study 2.

| Zone             | Center [m]         | Radius [m] | Science | Terrain |
|------------------|--------------------|------------|---------|---------|
| LZ0 Primary      | (16.7, 12.2, 0.85) | 0.90       | 0.95    | 0.86    |
| LZ1 South        | (16.2, 2.4, 0.85)  | 0.86       | 0.77    | 0.90    |
| LZ2 North Backup | (13.6, 12.5, 0.85) | 0.84       | 0.86    | 0.82    |
| LZ3 Mid-route    | (8.8, 1.8, 0.85)   | 0.82       | 0.63    | 0.76    |

## 11.2 Primary rollout parameters

**Table 11.2:** Primary parameters of the contingency-aware rollout.

| Quantity         | Value          | Quantity                    | Value              |
|------------------|----------------|-----------------------------|--------------------|
| Simulation step  | 0.05 s         | HOCBF $\alpha_1 = \alpha_2$ | 0.5                |
| Nominal $K_p$    | 0.95           | Nominal $K_d$               | 1.35               |
| Contingency      | $r = 2, p = 4$ | Gain $\gamma_c$             | 0.18               |
| Cost of $\omega$ | 0.10           | $\omega_{\max}$             | 80                 |
| Primary forcing  | guidance       | Primary solver              | conjugate gradient |
| LZ0 failure time | 18 s           | Approach altitude           | 4.2 m              |



Primary site is deliberately invalidated during flight; science scores are planning metadata, not safety certificates.

**Figure 11.1:** Analytic 3-D world containing towers, a gate structure, suspended obstacles, a central spire, terrain features, and four candidate landing zones. LZ0 is deliberately invalidated during the flight.

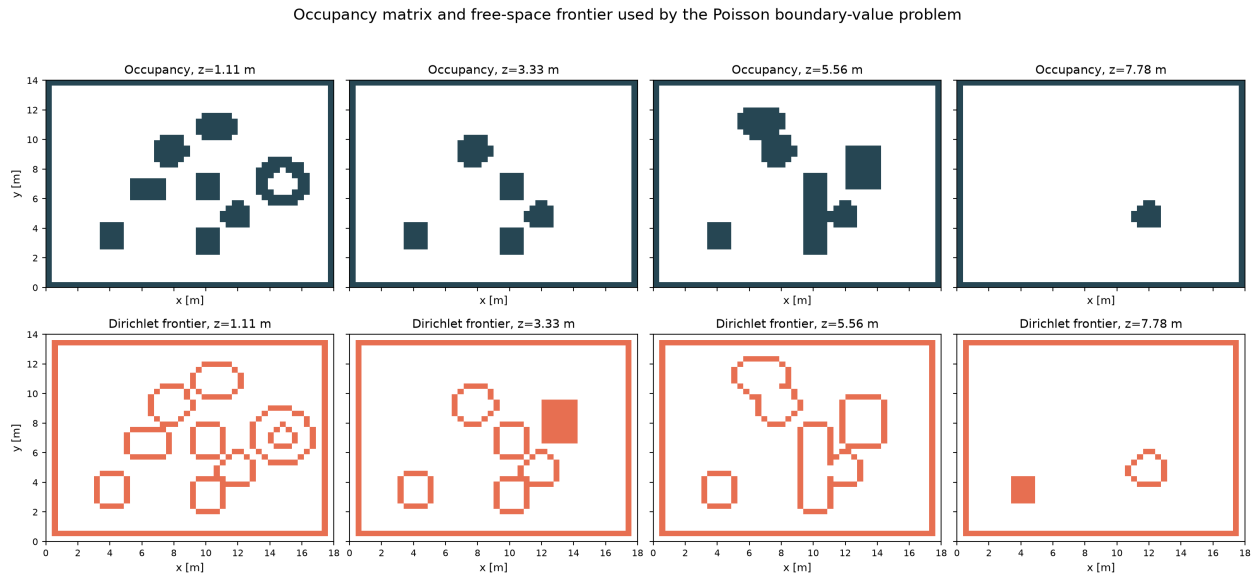
# Three-Dimensional Poisson Construction

## 12.1 Voxelization and obstacle inflation

The analytic obstacles are rasterized into voxels. A geometric inflation of 0.22 m is applied, and the complete outer boundary is marked occupied. The supplied run contains:

- 11018 occupied voxels;
- 40054 free voxels;
- 9333 frontier voxels;
- 30721 interior unknowns;
- 204849 nonzero entries in  $A$ .

Figure 12.1 should be read row by row. The upper row shows occupancy at several altitudes, while the lower row shows free cells whose field value is prescribed as zero. Near  $z = 7.78$  m, only the tallest structures remain. Near  $z = 1.11$  m, terrain obstacles and the crater rim become visible.



**Figure 12.1:** Horizontal occupancy and Dirichlet-frontier slices at four altitudes. Geometry changes with  $z$ : some obstacles disappear, some are suspended, and the outer wall remains a Dirichlet boundary at every level.

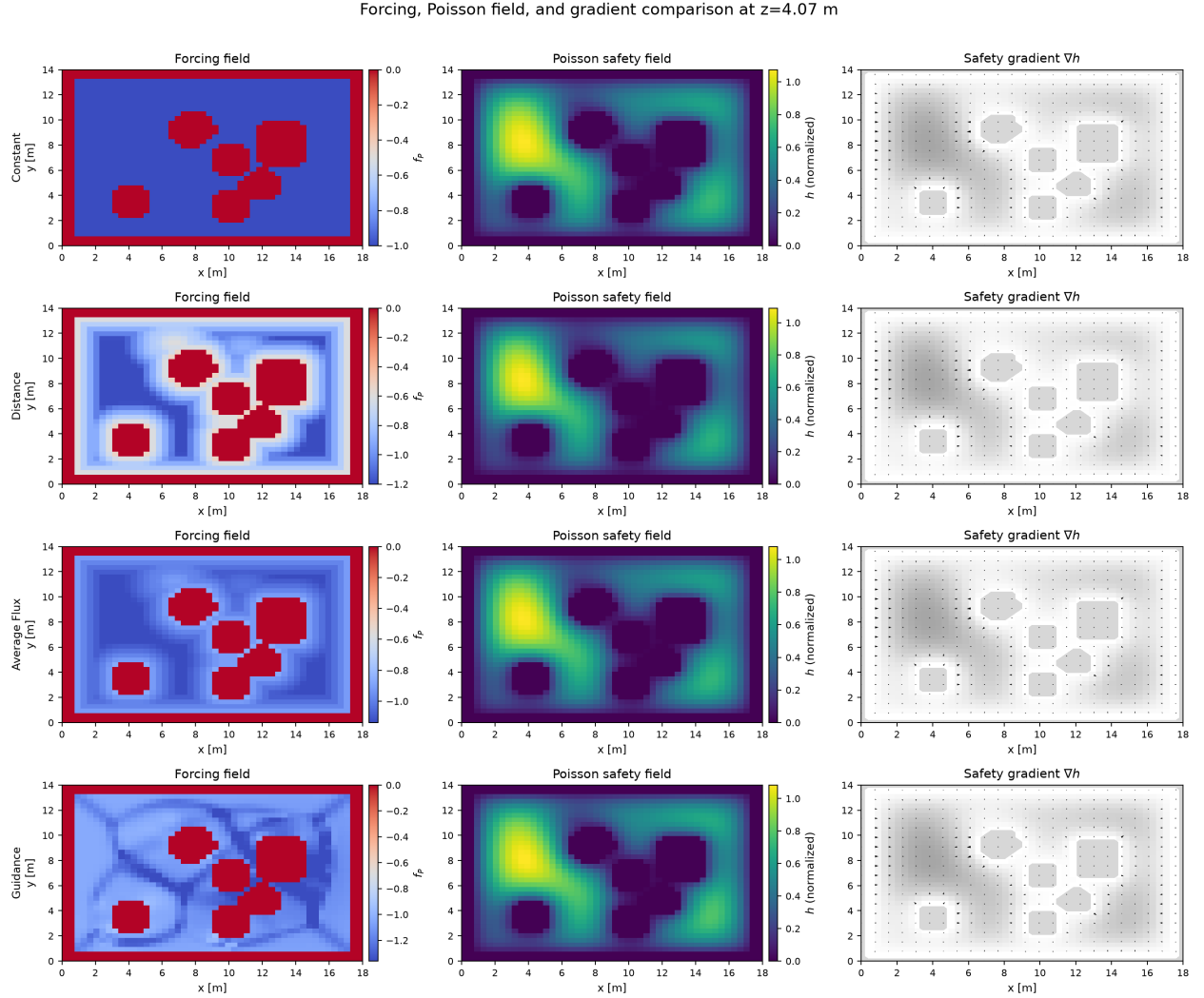
## 12.2 Forcing, normalized field, and gradient

The script normalizes the raw Poisson solution using the 99th percentile of interior values and defines

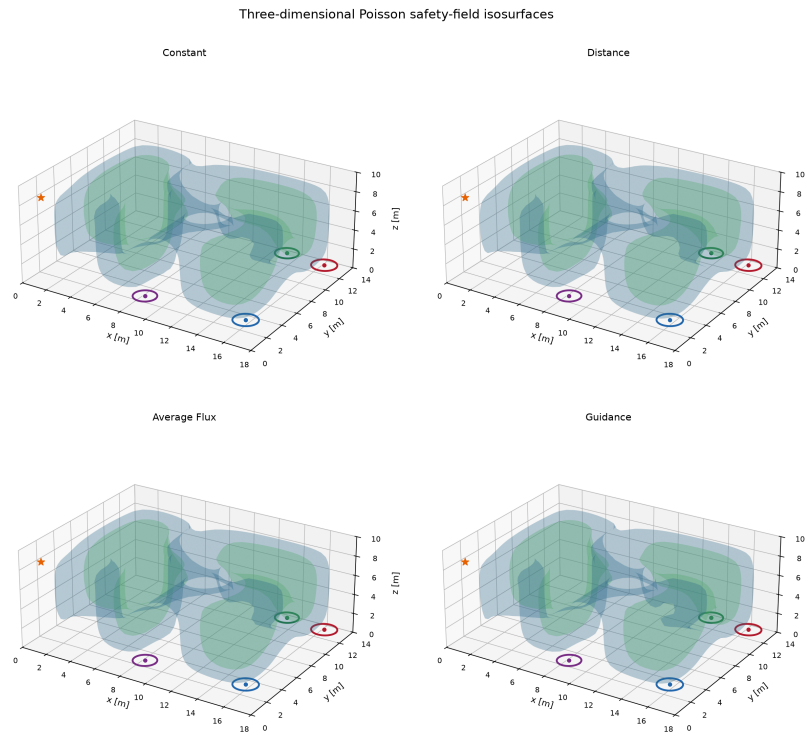
$$h_{\text{CBF}} = h_{\text{norm}} - 0.003. \quad (12.1)$$

Derivatives are computed from a field smoothed with a Gaussian filter of approximately 0.62 grid cells, and the Hessian is numerically clipped. These choices stabilize the reduced-order prototype, but they are part of the numerical approximation and must be documented as such.

In Figure 12.2, the obstacle domain is identical in every row. Only the forcing changes. Therefore, the differences in  $h$  and  $\nabla h$  are attributable to the interior PDE shaping rather than to a different obstacle map.



**Figure 12.2:** Comparison near  $z = 4.07$  m. Each row shows the forcing, normalized Poisson field, and safety gradient for one forcing family. The occupancy is unchanged; the forcing changes the interior field and therefore the HOCBF corrections.

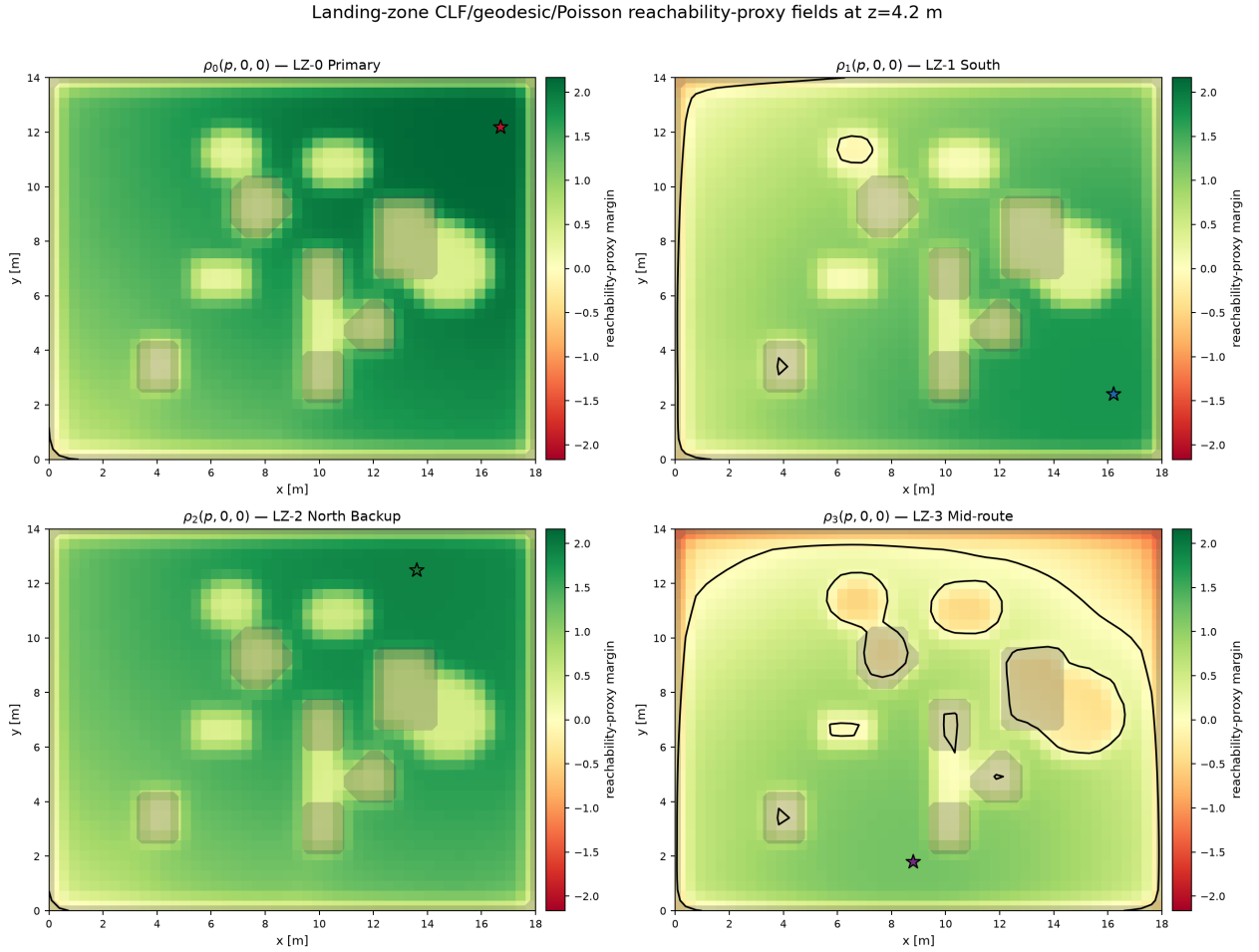


**Figure 12.3:** Two Poisson isosurface levels for each forcing field. The level surfaces surround broad free regions and narrow around corridors and complex obstacles.

# Landing Certificates and Combinatorial Geometry

## 13.1 Per-zone $\rho_i$ fields

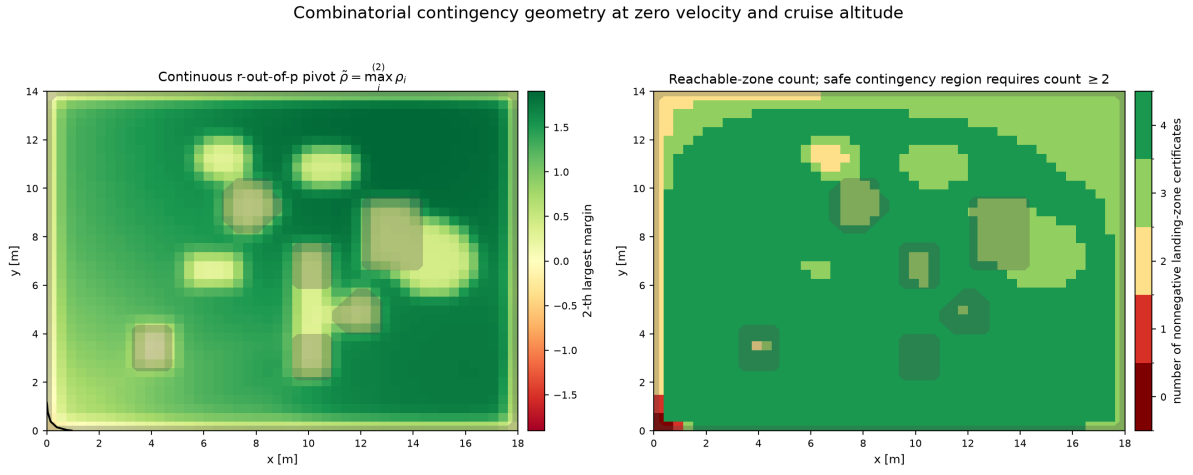
The maps in Figure 13.1 are computed at zero velocity and initial time so that the six-dimensional certificate can be displayed in a two-dimensional slice. During the actual simulation,  $W_i$  also depends on velocity, time, energy consumption, and blockage. Consequently, the figure does not represent a fixed reachability set in six-dimensional state space. It represents the section  $\rho_i(p, 0, 0)$  at cruise altitude.



**Figure 13.1:** Landing-zone proxies  $\rho_i(p, 0, 0)$  at cruise altitude. Green denotes positive margin, the black curves mark  $\rho_i = 0$ , and the star in each panel identifies the associated landing zone.

## 13.2 Pivot margin and number of available zones

The left panel preserves a continuous margin, while the right panel compresses the information into an integer count. A point with four positive certificates may lose two alternatives and still satisfy the



**Figure 13.2:** Left: the second-largest certificate because  $r = 2$ . Right: the number of nonnegative certificates. The contingency region requires a count of at least two.

$r = 2$  requirement. This flexibility is exactly what distinguishes an  $r$ -out-of- $p$  condition from the more conservative conjunction requiring all  $p$  zones to remain viable.

# Flight, Failure, and Safe Diversion

## 14.1 Nominal reference and auditable corridors

The nominal acceleration is

$$a_{\text{nom}} = K_p(p_{\text{ref}} - p) - K_d v. \quad (14.1)$$

The planner does not directly use the gradient of the geodesic-distance map to generate the reference. Instead, the script contains deterministic north and south corridor templates and a look-ahead reference point. This design reduces chatter near finite-difference saddle points and makes the nominal mission layer auditable. It is not a proof of globally optimal planning.

Descent is enabled only after the failure event, a dwell period following any switch, a horizontal proximity condition, and a horizontal speed condition. The vertical reference descends through a bounded look-ahead so that the HOCBF retains braking authority near the ground boundary.

## 14.2 Trajectory and switch event

Immediately after the failure is processed, the reported certificates are

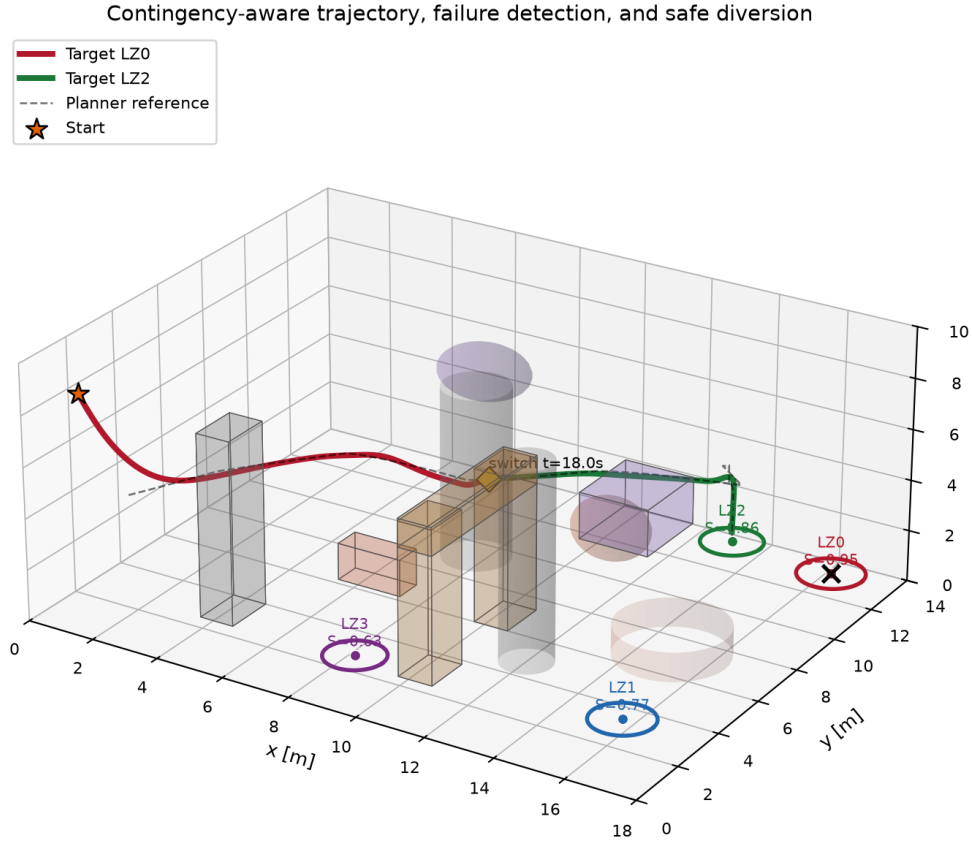
$$(\rho_0, \rho_1, \rho_2, \rho_3) = (-0.75, 0.966, 1.532, 0.692). \quad (14.2)$$

LZ0 is marked unavailable, and LZ2 receives the highest backup score. The switch does not require the environmental HOCBF to be disabled, nor does it require the Poisson field to be reinitialized.

## 14.3 Synchronized time histories

The figure should be interpreted panel by panel:

1. **Environmental safety:**  $h_{\text{CBF}}$  remains positive. The HOCBF residual approaches zero when the environmental row becomes active.
2. **Landing-zone proxies:**  $\rho_0$  drops when the primary site is declared blocked.  $\rho_2$  remains high and becomes the active zone.
3. **Combinatorial margin:**  $\tilde{\rho}$  never crosses zero; its minimum is approximately 0.389.
4. **Reachable-zone count:** the count decreases from four to three and later to two, but it never falls below the required  $r = 2$ .
5. **Target selection:** the active index changes from 0 to 2 at the failure event.
6. **QP intervention:** correction increases near obstacles and during contingency reorganization. The emergency slack remains zero.



**Figure 14.1:** Primary trajectory. Red denotes the segment targeting LZ0 before failure. The diamond marks the switch at  $t = 18$  s. Green denotes diversion toward LZ2 and the final descent.

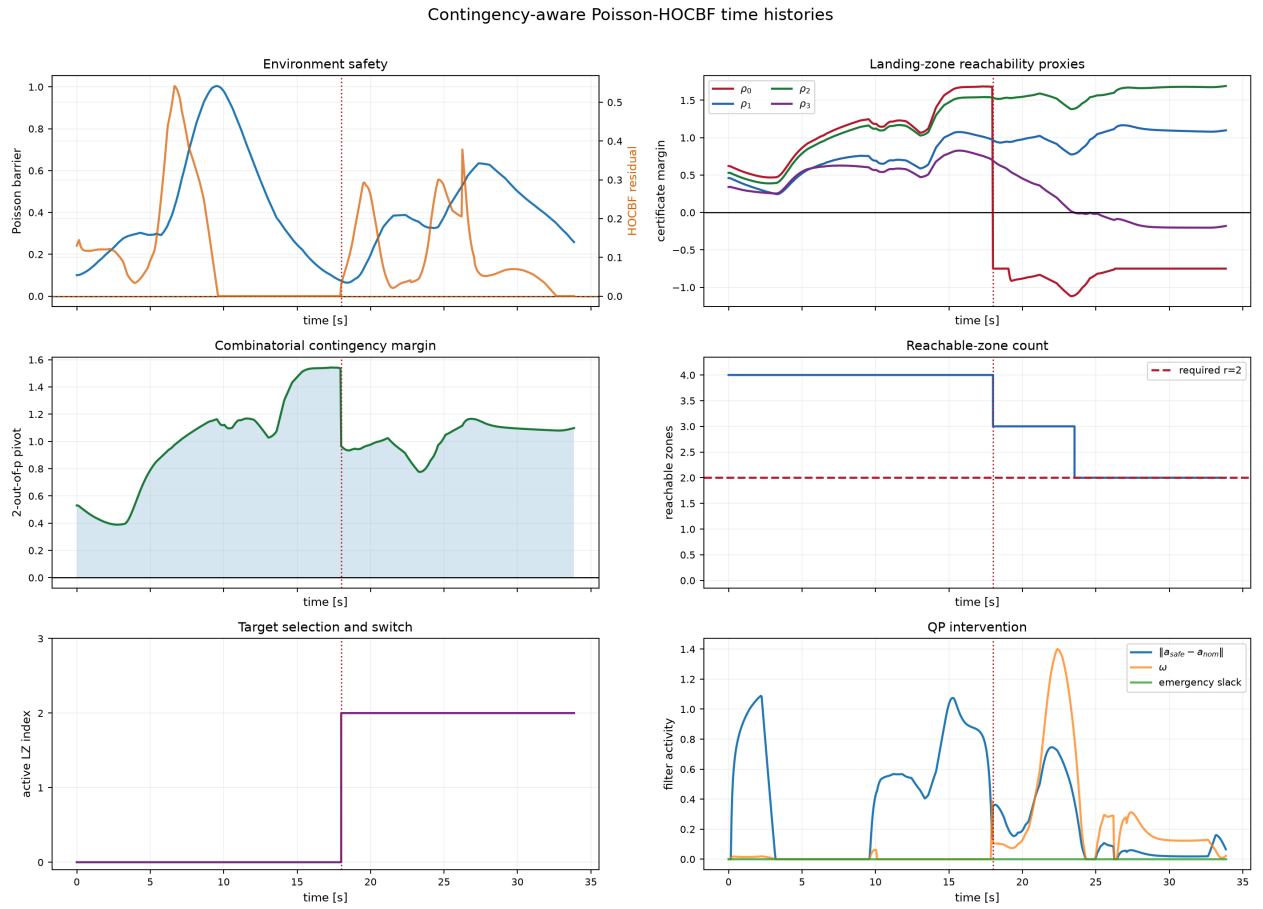
**Table 14.1:** Primary results of Case Study 2 with  $\alpha = 0.5$  and guidance forcing.

| Metric                                         | Value   | Metric                       | Value                   |
|------------------------------------------------|---------|------------------------------|-------------------------|
| Outcome                                        | landed  | Final zone                   | LZ2                     |
| Duration                                       | 33.85 s | Steps                        | 678                     |
| Collision                                      | no      | Contingency loss             | no                      |
| $\min h_{\text{CBF}}$                          | 0.0643  | $\min \text{HOCBF residual}$ | $-7.89 \times 10^{-17}$ |
| $\min \tilde{\rho}$                            | 0.3891  | Minimum count                | 2                       |
| Final distance                                 | 0.422 m | Final speed                  | 0.380 m/s               |
| Switches                                       | 1       | Emergency slack              | 0                       |
| $\int \ a_{\text{safe}} - a_{\text{nom}}\  dt$ | 10.814  | $\max \omega$                | 1.401                   |

## 14.4 Metrics of the primary rollout

### Observed result

The rollout preserved the proxy-defined 2-out-of-4 set, avoided discrete occupancy, switched away from the blocked primary zone, and terminated within the configured landing tolerances of LZ2. No emergency slack was used, and the active CLF did not need to be removed in the primary run.



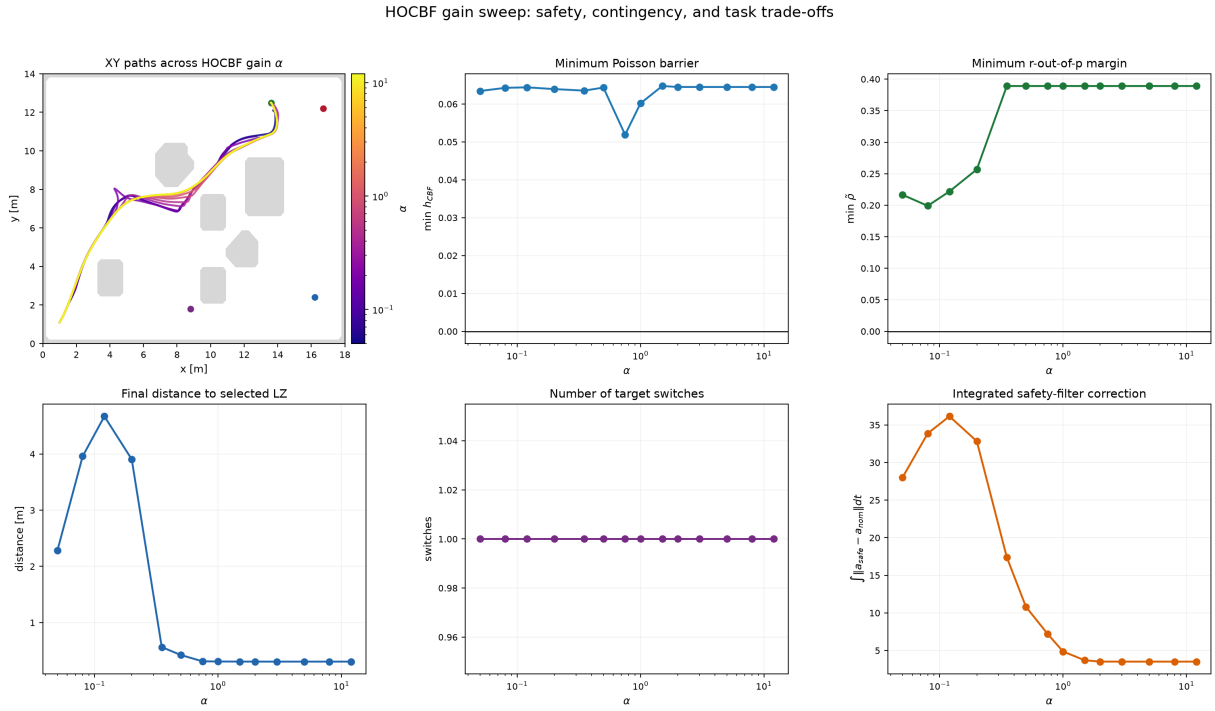
**Figure 14.2:** Six synchronized diagnostics. The vertical red line indicates the LZ0 failure event at  $t = 18$  s.

# Sensitivity Studies for Case Study 2

## 15.1 Sweep of the HOCBF gain

For  $\alpha = 0.05, 0.08, 0.12$ , and  $0.2$ , the simulation does not land before the maximum horizon and requires emergency slack. Beginning near  $\alpha = 0.5$ , the primary rollout lands without slack; for  $\alpha \geq 1.5$ , the reported metrics are nearly unchanged in this specific scene.

This does not imply that a larger  $\alpha$  is universally better. In a relative-degree-one CBF, a larger gain permits faster decrease of  $h$  away from the boundary. In the HOCBF, the gain appears in several terms and interacts with velocity, curvature, actuator saturation, and sample time. The following figure combines path shape, safety, contingency, destination, and intervention.



**Figure 15.1:** Sweep of  $\alpha$  from 0.05 to 12. The gain changes intervention, arrival time, and bounded-input numerical feasibility.

## 15.2 Comparison of forcing fields

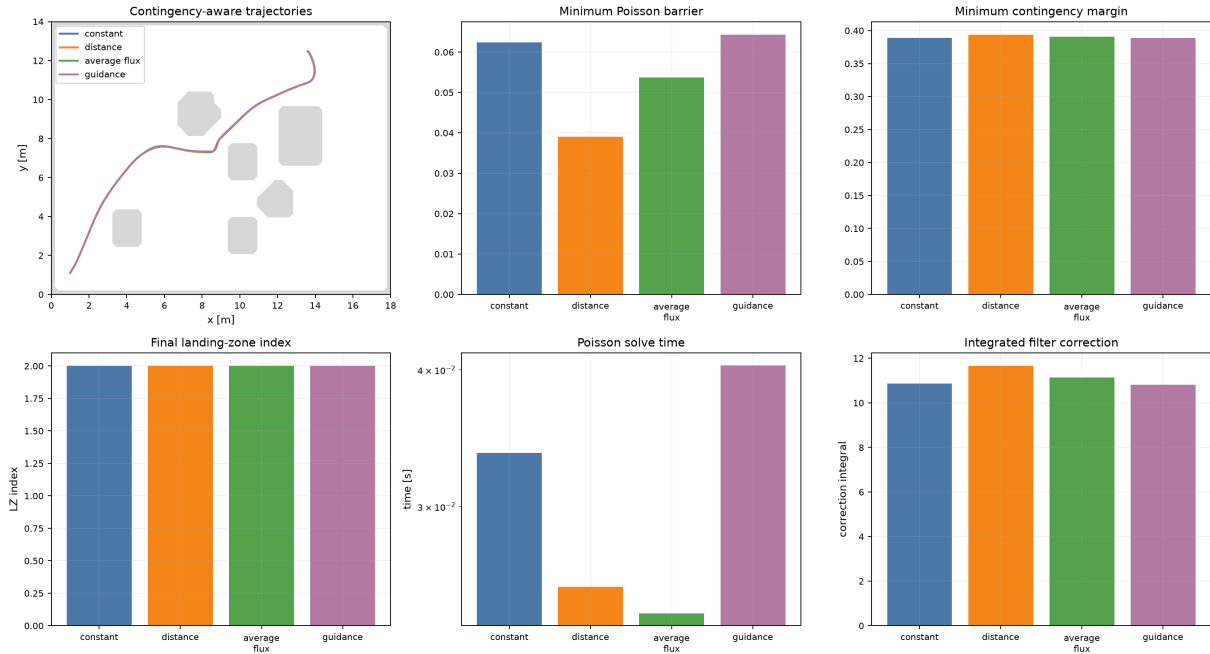
**Table 15.1:** Forcing-field comparison in Case Study 2.

| Forcing      | Duration [s] | $\min h_{\text{CBF}}$ | $\min \tilde{\rho}$ | Integrated correction |
|--------------|--------------|-----------------------|---------------------|-----------------------|
| constant     | 33.95        | 0.062 4               | 0.388 7             | 10.86                 |
| distance     | 34.60        | 0.039 0               | 0.393 9             | 11.66                 |
| average flux | 34.15        | 0.053 7               | 0.390 4             | 11.14                 |
| guidance     | 33.85        | 0.064 3               | 0.389 1             | 10.81                 |

All four runs land and preserve the configured contingency set, but the environmental margin and

accumulated correction change. The visual comparison must be read together with the table: a slightly shorter path does not necessarily imply a more robust or better-conditioned safety field.

Poisson forcing-function comparison under the contingency-aware filter



**Figure 15.2:** Effect of the forcing choice on trajectory and closed-loop metrics. Every case lands and preserves contingency, but environmental margin and integrated correction differ.

### 15.3 Comparison of sparse solvers

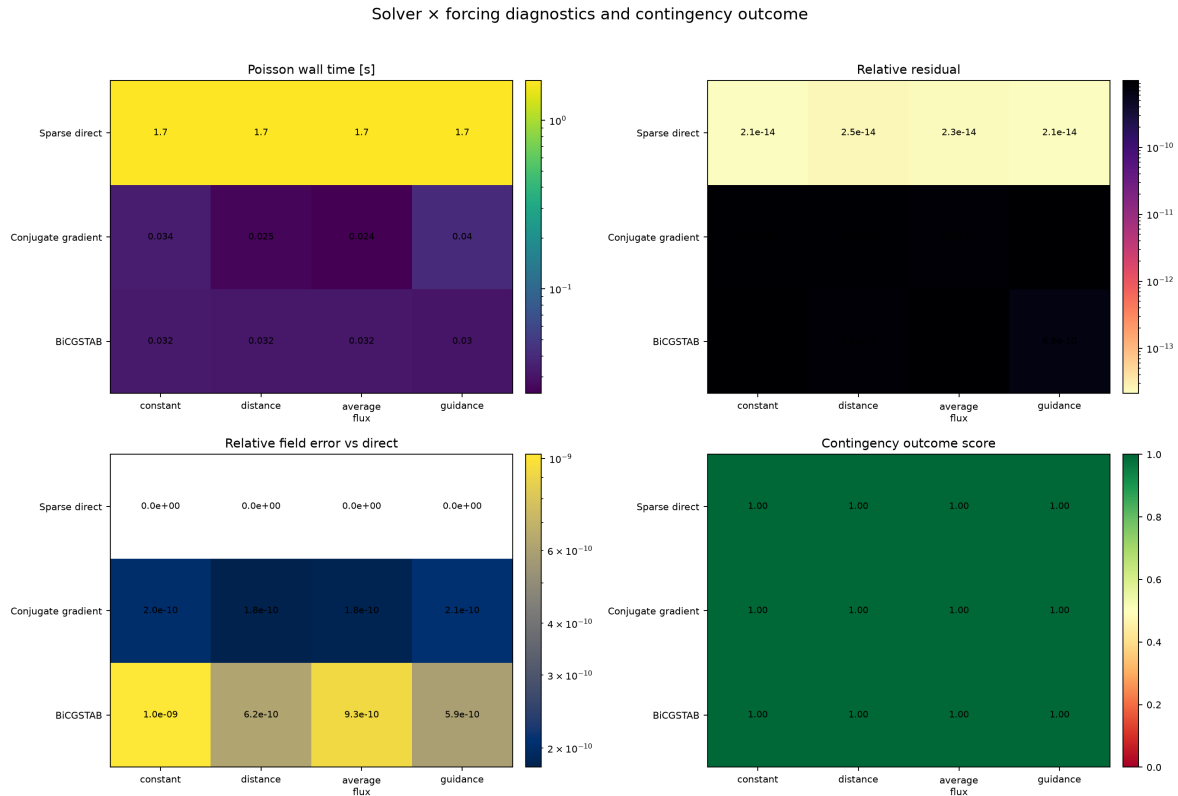
For the supplied execution:

- sparse direct requires approximately 1.7 s per field and reaches residuals near  $10^{-14}$ ;
- conjugate gradient requires approximately 0.024–0.040 s and reaches relative residuals near  $10^{-9}$ ;
- BiCGSTAB requires approximately 0.030–0.032 s;
- relative field errors against the direct solution lie between  $10^{-10}$  and  $10^{-9}$ ;
- all twelve closed-loop rollouts obtain outcome score 1.0 and terminate at LZ2.

These times depend on the processor, BLAS implementation, sparsity pattern, tolerance, and grid size. They are not universal benchmarks. The heatmap separates solve time, residual quality, field equivalence, and closed-loop outcome.

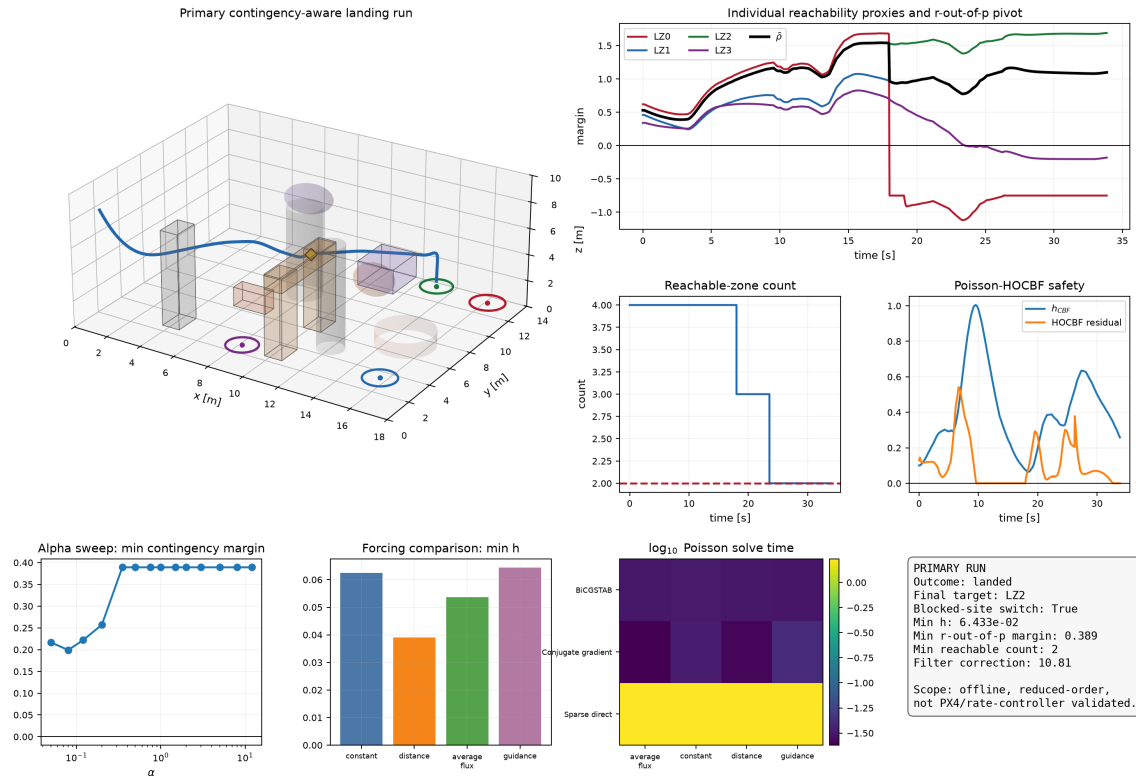
### 15.4 Integrated dashboard

The dashboard does not introduce a new proof. It combines the primary trajectory, all landing certificates, the  $r$ -out-of- $p$  count, Poisson-HOCBF safety, and the three sensitivity studies on one page. Its value is operational: it makes it possible to identify whether an apparent timing improvement was obtained at the cost of reduced margin, nonzero slack, or excessive safety-filter correction.



**Figure 15.3:** Solver-by-forcing comparison: time, algebraic residual, field error relative to sparse direct, and closed-loop outcome. On this grid, CG and BiCGSTAB are substantially faster than the direct solve and reproduce the same rollout outcome.

Poisson-CBF contingency landing study — integrated dashboard



**Figure 15.4:** Integrated summary of the study: primary trajectory, landing certificates, reachable-zone count, HOCBF diagnostics, gain sweep, forcing comparison, solver comparison, and primary metrics.

## Part IV

# Interpretation, Guarantees, and Extensions

# Comparison of the Two Case Studies

**Table 16.1:** Conceptual differences between the two case studies.

| Aspect                    | Case Study 1: static image         | Case Study 2: 3-D contingency         |
|---------------------------|------------------------------------|---------------------------------------|
| Geometric input           | rectified and segmented photograph | voxelized analytic obstacles          |
| Poisson dimension         | 2-D                                | 3-D                                   |
| Reduced model             | $\dot{p} = u$                      | $\dot{p} = v, \dot{v} = a$            |
| Environmental certificate | velocity-level CBF                 | acceleration-level HOCBF              |
| Task                      | one fixed goal point               | one active zone among four            |
| Contingency               | none                               | preservation of 2 out of 4 zones      |
| QP structure              | projection onto one half-space     | HOCBF + CLF + four combinatorial rows |
| Main result               | nominal collides; CBF reaches goal | LZ0 fails; switch to LZ2 and land     |

The first example isolates the essential connection

$$O \longrightarrow h \longrightarrow \nabla h \longrightarrow u. \quad (16.1)$$

The second adds velocity, curvature, acceleration, actuator limits, multiple targets, resource depletion, and target-switching logic. This extra complexity does not change the fundamental role of Poisson's equation: it converts a complex geometric boundary into one unified differentiable field.

# What Is Guaranteed and What Is Only Observed

**Table 17.1:** Hierarchy of claims supported by the current artifacts.

| Claim                            | Available support                        | Limitation                                                   |
|----------------------------------|------------------------------------------|--------------------------------------------------------------|
| $A\mathbf{h} \approx \mathbf{b}$ | measured algebraic residual              | depends on grid and solver tolerance                         |
| $h = 0$ on the frontier          | numerical boundary check                 | discrete frontier, not an exact continuous surface           |
| CBF/HOCBF inequality             | measured QP residual                     | reduced model and continuous-time theorem                    |
| No collision in rollout          | discrete occupancy check                 | no real tracking or uncertainty model                        |
| $r$ -out-of- $p$ preservation    | $\tilde{\rho} > 0$ for numerical proxies | proxies are not HJ reach-avoid values                        |
| Real-vehicle landing             | not demonstrated                         | requires PX4, estimation, actuation, and robustness analysis |

## Technical warning

A smooth heatmap is not a safety guarantee. A guarantee requires a function with reliable derivatives, a feasible inequality, a model that correctly describes how the actual input enters the dynamics, a sufficiently regular feedback law, and an implementation that respects sampling, latency, saturation, and state-estimation uncertainty.

# Extension to Video and Dynamic Maps

The supplied vision repository also contains webcam, phone-stream, and video workflows. Its architecture separates frame capture, perception, and a background Poisson worker. The queue has capacity one: a newly available map can replace an older task that is still waiting, and a stale result is not allowed to overwrite the newest valid field.

Each live field is currently quasi-static:

$$\Delta h_t = f_{P,t} \tag{18.1}$$

for one occupancy snapshot. A formal guarantee with moving obstacles would require a time-varying barrier condition such as

$$\frac{\partial h}{\partial t} + \nabla h^\top f(x, u) \geq -\alpha(h), \tag{18.2}$$

together with a moving-boundary model and explicit bounds on perception error, field age, latency, and tracking. The finite-difference estimate of  $h_t$  shown by the live dashboard is a diagnostic, not a proof.

# Recommended Validation Ladder

A defensible development sequence is:

1. repeat residual, boundary, gradient, and Hessian validation under grid refinement;
2. inject occupancy noise and pose error, and determine robust geometric inflation;
3. verify bounded-input QP feasibility under realistic velocity and acceleration limits;
4. connect ROS 2, Gazebo, and PX4 SITL through Offboard mode;
5. measure tracking error between the reduced-order command and the simulated vehicle;
6. robustify the CBF against tracking, estimation, and latency bounds;
7. validate on Crazyflie with OptiTrack;
8. migrate to an instrumented X500-class platform;
9. replace or compare the proxies  $\rho_i$  with Hamilton–Jacobi reach–avoid values when dimensionality permits.

# Conclusions

Poisson’s equation provides a mathematically clean interface between perception and control. Binary occupancy is discontinuous and does not directly provide a local control direction. The Poisson solve transforms that occupancy geometry into a field  $h$  with a gradient and Hessian. A velocity-level CBF uses  $\nabla h$  to project commands. An acceleration-level HOCBF additionally uses  $\nabla^2 h$  to account for curvature and vehicle velocity. The contingency layer introduces one certificate per landing zone and preserves the  $r$ -th largest certificate through  $p$  smooth inequalities.

The two examples demonstrate different levels of maturity. The first clearly shows that the nominal controller collides while the Poisson-CBF projection reaches the goal. The second shows, within an offline double-integrator prototype, that Poisson, HOCBF, CLF, actuator bounds, and a failure event can be combined to switch from LZ0 to LZ2 without losing the proxy-defined 2-out-of-4 margin. The next step is not merely to place the same script inside PX4. It is to close the gaps in model fidelity, perception, latency, tracking, and robustness that separate a reduced-order numerical demonstration from an experimental safety guarantee.

# Compact Derivations

## A.1 Projection onto a half-space

Consider

$$\min_u \frac{1}{2} \|u - u_0\|^2 \quad \text{subject to} \quad g^\top u \geq b. \quad (\text{A.1})$$

The Lagrangian is

$$\mathcal{L}(u, \lambda) = \frac{1}{2} \|u - u_0\|^2 + \lambda(b - g^\top u), \quad \lambda \geq 0. \quad (\text{A.2})$$

Stationarity gives

$$u = u_0 + \lambda g. \quad (\text{A.3})$$

If the inequality is active,

$$\lambda = \frac{b - g^\top u_0}{\|g\|^2}, \quad (\text{A.4})$$

which yields the closed-form correction used in Case Study 1.

## A.2 Gradients of the quadratic contingency term

Partition

$$P = \begin{bmatrix} P_{pp} & P_{pv} \\ P_{vp} & P_{vv} \end{bmatrix}, \quad e_i = \begin{bmatrix} p - \ell_i \\ v \end{bmatrix}. \quad (\text{A.5})$$

Then

$$\nabla_p \left( \frac{e_i^\top P e_i}{s_i} \right) = \frac{2}{s_i} (P_{pp}(p - \ell_i) + P_{pv}v), \quad (\text{A.6})$$

$$\nabla_v \left( \frac{e_i^\top P e_i}{s_i} \right) = \frac{2}{s_i} (P_{vp}(p - \ell_i) + P_{vv}v). \quad (\text{A.7})$$

The kinetic term contributes

$$\nabla_v \left( w_v \frac{\|v\|^2}{v_{\max}^2} \right) = \frac{2w_v}{v_{\max}^2} v. \quad (\text{A.8})$$

## A.3 Connection to Hamilton–Jacobi reachability

For the double-integrator model

$$\dot{p} = v, \quad \dot{v} = a, \quad (\text{A.9})$$

a reachability value function  $J_i(p, v, t)$  would satisfy a first-order Hamilton–Jacobi PDE of the form

$$\frac{\partial J_i}{\partial t} + \nabla_p J_i^\top v + \min_{a \in \mathcal{A}} \nabla_v J_i^\top a = 0, \quad (\text{A.10})$$

with target and avoid-set boundary conditions. Under the convention  $J_i \leq 0$  for reachable states, one could define

$$\rho_i = -J_i. \tag{A.11}$$

The combinatorial layer could then preserve the  $r$ -th largest reachability margin. The current code does not solve this PDE; it uses the tractable proxy in (5.2).

# Reproduction Commands

## B.1 Python environment

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install -e ../poisson_safety_box
pip install -e ../cbf_safety_box
pip install -e "vision_poisson_experiments[dev]"
```

## B.2 Static image experiment

```
python 01_static_image_poisson_cbf/run_experiment.py \
  --image examples/assets/image1.png \
  --config 01_static_image_poisson_cbf/config.yaml \
  --output outputs/static_run \
  --assume-top-down --headless
```

## B.3 Contingency-aware landing experiment

```
python run_poisson_cbf_contingency_landing_study.py \
  --output-dir outputs/contingency_poisson_landing \
  --grid-shape 48,38,28 \
  --r-contingency 2 \
  --num-landing-zones 4 \
  --fixed-forcing guidance \
  --solver conjugate_gradient \
  --max-speed 2.0 \
  --max-acc 1.35
```

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